银湾研究院（GreyBay Institute）
Great Bay Area Innovation and Research Center

Risk Factors of Underweight and Interventions to Prevent or Mitigate Underweight in Young Adult Females in Japan: An Evidence-based Research Project for China

EDITOR(S): *Hana Hayashi*

CONTRIBUTOR(S): *Yue Li, Shujuan Chen, Akihiro Shimoda*

REVIEWER(S): *Min Zhang, Wei Li*

PUBLISHED: Nov 14th, 2025

REPORT NO.: GBI-HP-2025-02

SERIES: GreyBay Health Policy Report Series

INSTITUTION: GreyBay Institute, Shenzhen, China

This report was supported by the National Social Science Fund of China (NSSFC) and the National Natural Science Foundation of China (NSFC). The funding bodies had no role in the design of the study, analysis and interpretation of the evidence, or the preparation of the report.

© 2025 GreyBay Institute

This report is licensed under Creative Commons Attribution 4.0 (CC BY 4.0).

Abstract

Underweight among young women has become a notable public health concern in several East Asian societies. In Japan, more than 20% of women in their twenties are underweight, largely influenced by strong sociocultural preferences for thinness. This condition is associated with significant health risks, including eating disorders, osteoporosis, reproductive complications, and increased mortality. Understanding the drivers of underweight and identifying effective prevention strategies is therefore important for countries facing similar trends.

This report, produced by the GreyBay Institute, synthesizes existing evidence on the risk factors and potential interventions related to underweight among young adult women, using Japan as a key reference case and exploring its relevance for China. Based on a review of the literature, a hierarchical ecological framework is developed to organize determinants into social and cultural, family, and individual levels. Within the individual level, psychological, behavioral, sociodemographic, and physiological factors are examined, highlighting the central role of thin-ideal norms, media influence, body dissatisfaction, and weight-control behaviors in the development of underweight.

The report also reviews empirical studies evaluating interventions, including body-positive media exposure, cognitive dissonance programs, mindfulness training, nutrition education, and exercise-based approaches. Evidence suggests that interventions targeting psychological and cognitive factors—particularly body image and internalization of thin ideals—show promising effects in reducing eating disorder risk and unhealthy weight control behaviors.

By drawing lessons from Japan's experience, this report provides an evidence-based foundation for future research and policy development in China. It aims to inform strategies that promote healthier body image, prevent underweight, and support the well-being of young women in rapidly changing social environments.

Table of Contents

1. BACKGROUND **4**

2. RISK FACTORS OF UNDERWEIGHT **7**

2.1 OVERVIEW **7**

2.2 SOCIAL AND CULTURAL LEVEL **8**

2.3 FAMILY LEVEL **11**

2.4 INDIVIDUAL LEVEL **13**

2.4.1 PSYCHOLOGICAL FACTORS 13

2.4.2 BEHAVIORAL FACTORS 16

2.4.3 SOCIODEMOGRAPHIC FACTORS 16

2.4.4 PHYSIOLOGICAL FACTORS 17

3. INTERVENTIONS TO PREVENT OR MITIGATE UNDERWEIGHT **17**

3.1 OVERVIEW **17**

3.2 LIST OF SUCCESSFUL INTERVENTION **18**

4. FINDINGS AND CONCLUSIONS **33**

REFERENCES **36**

1. Background

Decrease in BMI of young female is a tendency observed worldwide by typically in Asian culture. It is without question that over the 35 years and even until now in 2020, adolescents and young adult female are getting thinner. The findings of a study were shown in Figure 1-1 and Figure 1-2. I could be seen that young women in Japan and Korea have generally lower body mass index (BMI; weight [kg]/height^2 [m^2]) compare to those in the US. The difference further increase in higher age intervals after 18 years old. In Japanese society particularly, it was shown that from 1960 to 1995, females aged from 15~24 years old increased in height and weight but their BMI dropped from 21.5 to 20.5. [1] Asian young women, especially Japanese women, have been reported to strongly desire to be thin, even though they have a lower BMI than other ethnic groups. [2] As a result, more than 20% of women in their 20s are underweight (BMI <18.5 kg/m^2) in Japan, a percentage much higher than in most developed countries. [3] On average, the perceived ideal weight was 5.2 kg lower than the current weight. Young women have a tendency to misunderstand their body shape and to regard themselves as heavier than they really were. [4]

Why underweight is a problem? The consequences of being underweight is sever and often neglected.

The biggest concern is eating disorders. There are two subtypes of eating disorders, bulimia nervosa and anorexia nervosa. Bulimia nervosa is characterized by binge eating and purging, as well as excessive evaluation of one's self-worth in terms of body weight or shape. Purging can include self-induced vomiting, over-exercising, and the use of diuretics, enemas, and laxatives. Anorexia nervosa is characterized by extreme food restriction and excessive weight loss, accompanied by the fear of being fat. [5] Eating disorders cause sever health outcome. Anorexia nervosa increases the chance of getting osteoporosis. Thinning of the hair as well as dry hair and skin are also very common. The muscles of the heart will also start to change if no treatment is inflicted on the patient. This causes the heart to have an abnormally slow heart rate along with low

blood pressure. Bulimia symptoms include heart problems like an irregular heartbeat that can lead to heart failure and death may occur. The probability of a gastric rupture increases. Due to laxative abuse, irregular bowel movements may occur along with constipation. [6] From the facts above, we can see that both types of eating disorders are highly related to underweight. We noticed many studies were focusing on eating disorders, but they did not further take underweight as the next consequence due to the fact that underweight is a well-known symptom of eating disorders. [7] Though usually eating disorders are considered as an upstream cause of underweight or a comorbidity of underweight, we found it inadequate to ignore eating disorder in this setting. The reasons are two-folded, (1) complications of underweight could certainly worsen the consequence of eating disorders, and (2) there are significant overlapping on the risk factors.

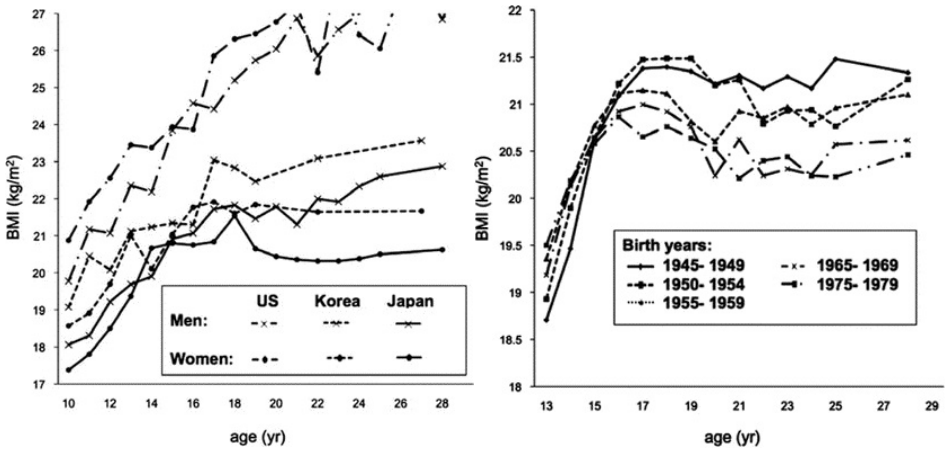

Fig. 1-1 Changes in BMI of young men and women with age in US, Korea and Japan

Fig. 1-2 Changes in BMI of young people with age in Japan stratified by birth year

There are other health issues of underweight other than eating disorders. According to a study in 2016, being underweight increases a woman's risk of osteoporosis, which is where the bones are brittle and more prone to breaking. [8] A study published in An International Journal of Obstetrics & Gynecology revealed that a woman who is pregnant and underweight is at a higher risk for pre-term labor, which means having a baby before 37 weeks. [9] According to a study published in the journal

BMC Public Health, being underweight is associated with an increased risk for mortality when compared to people with an average BMI. The researchers suggested that being underweight may impair a person's healing processes following an accident or trauma compared to a person with an average BMI. [10] Other health effects include having skin, hair, or teeth problems, getting sick frequently, feeling tired all the time, slow and impaired growth or premature births. [11]

However, to our current knowledge, although there are abundant studies showing risk factors that leads to underweight, no study has been conducted systematically reviewing those factors and helping us understand the underlying relationships between them. Neither do we have the knowledge on the hierarchy of those rick factors. Which acts more like a fundamental cause, which factors are derivatives of that cause? What are the characteristics of young women that are highly vulnerable? Do the risk factors have interactions with each other that lead to moderation or intensification of the effects? As for ways to prevent or mitigate underweight in young women, what are the interventions possible to tackle the problem? Do they have different intervention effects comparing with each other? We need a thorough research to answer all these questions.

This chapter of report have two aims, which are (1) identifying the risk factors that cause underweight in young women, and (2) searching for interventions that proved to have effect on preventing or mitigating the problem. As we were carrying out our research, we noticed that the cause of underweight is extremely complex, and it was not reasonable to include evidence that only linked to underweight directly. Thus, we included a variety of keywords into our searching formula that might be associated with underweight through our preliminary research. It also needed to be mentioned that not all included studies have a causal conclusion between the exposures and outcomes, we included as many studies with good qualities as possible because they help with our understanding of the relationships between the recognized factors.

2. Risk factors of underweight

2.1 Overview

Based on our research, we classified all factors that cause underweight in young women into three levels, social and cultural level, family level, and individual level. For individual level, the factors were further separated into four aspects, namely, psychological, behavioral, sociodemographic and physiological factors.

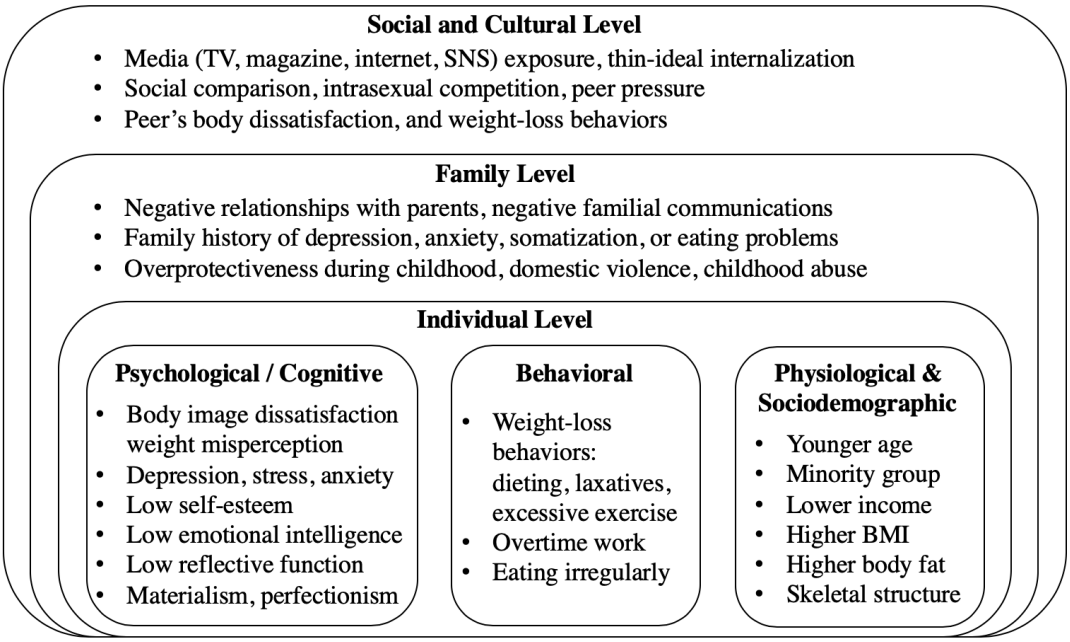

Fig. 2-1 An Ecological model for the hierarchy of factors that cause underweight in young women

Figure 2-1 depicts the hierarchy of the factors related to underweight in young women. Detailed factors are provided under each big category and will be further discussed in the next three subsections in accordance with the three levels. While in Figure 2-2, the logic web represents the hypothesized causal relationship between each factor, which was summarized from all review literatures. What need to be mentioned is that a large proportion of studies were association study that could not elude a casual conclusion due to study design, some of the links in this logic web might need to be further studied, but this at least helps to understand the underlying relationships

between the complex factors. Evidence that support each of the arrows are discussed in the corresponding subsections.

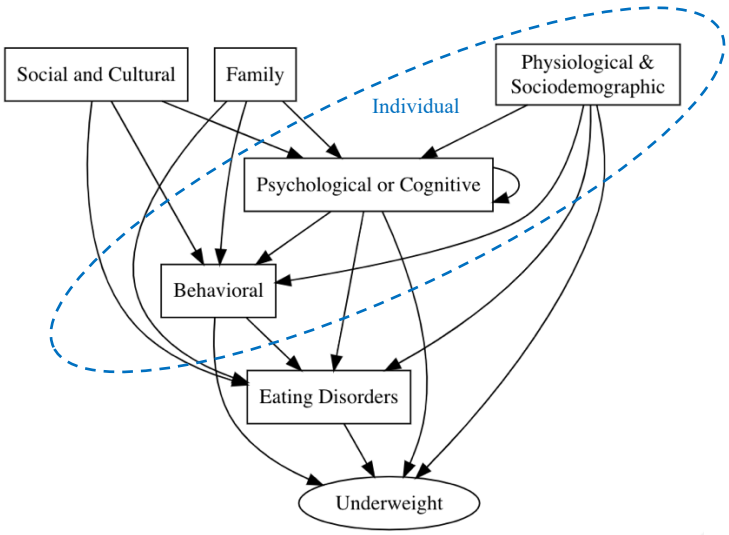

Fig. 2-2 The underlying logic web that lead to underweight in young women

For the outcome of our report, we were targeting at underweight at first, but we later added eating disorders as well. The rationale for this was stated in the introduction part. In a word, we took these two consequences as separate and paralleled outcomes in this report due to their similarities and potential interaction effects.

2.2 Social and cultural level

Social and cultural factors are considered as the most fundamental underlying reason for underweight and is also the hardest to intervene. They shape young women's values, influence their personality, and modify their psychological or cognitive beliefs. These could further lead to voluntary weight loss behaviors, which is a direct cause of eating disorder and underweight. Though there are obvious difference between eating disorder and underweight, many of the studies took eating disorder as the final outcome of interest. And considering the fact that eating disorder is strongly correlated with underweight, we included eating disorder in the searching formula as another outcome. In this subsection, direct consequences of social and cultural level factors that are supported by evidence will be discussed. Indirect consequences will be discussed

further in later sections.

Firstly, social and cultural environment is a determinant of young women's psychological and cognitive status. A study revealed that the greatest weight loss expectations of the participants came from mass media. [12] Media using underweight fashion model and celebrities increases drive for thinness, internalization of the ultra-thin ideal and body dissatisfaction. [13]–[23] This effect could be magnified in people with higher self-discrepancy (self-discrepancy: comparison between their "actual" self to internalized standards or the "ideal/ought self").[24] Inversely, when women processed mostly positive and rejected negative source information, their body investment decreased and body evaluation became more positive. [25] Interestingly, the effect of media tends to be stronger in population that matches the target population of the media. A study conducted on young Chinese women found that they were more sensitive to Chinese or Asian than Western mass media. [26] Similar culture-specific influencing patterns could also be seen in black and white communities. [27] Specifically, internet exposure [28], watching television [29], [30] (TV commercials using thin models [31], [32]), soaps and music television [33], [34] and reading magazines [35], [34] increased body dissatisfaction. After media exposure, internalization happens. It was revealed in one study that lower similarity that young women feels toward female models, or higher desire to look like the female model is associated with higher body surveillance. Romantic attachment was associated with increased body surveillance for male character. [36]

However, mass media is only one source of receiving the thin-ideal values, people whom young women having contact with might have higher power in shaping their thoughts. Social comparison, peer pressure, intrasexual competition [16], [37]–[41], social media comparison [42], peer negative communication [43], pressure from parents [44] all adds to the social and cultural environment and worsen the situation. Motivation from romantic partners also contribute to this [37], though females tend to be more dissatisfied of their bodies than their romantic partners actually perceived. [45] As a result, people surrounding young women pass their values and beliefs to them, letting them think that being thinner will make boy like them more, reducing weight

would lead to better life opportunities and greater acceptance from others. [44] Peers are strongly influencing roles, as is shown that perception of their peers' body dissatisfaction emerged as the strongest predictor of their own level of body dissatisfaction [30], [34], [46]. And healthy food consumption in young women was also proven to be influenced by peer's eating habit. [47]

Among the many ways of getting in touch with peers in everyday life, SNS is the most popular and thus is the potentially most influential path. Appearance-focused SNS use or active engagement with SNS photo activities, like FB [48] and Instagram [49], rather than overall SNS use, was related to body image concerns. [50], [51] Nevertheless, two studies pointed out that general SNS use is also associated with body dissatisfaction. [52] Profile maintenance, social grooming are related to lower weight satisfaction and greater drive for thinness. [40] More interestingly, not only viewing the content could cause anxiety or negative psychological impacts, females who post selfies to social media themselves also report feeling more anxious, less confident, and less physically attractive. [53] However, influences from peers do not always have to be negative. Emotional support could reduce the negative effect. [54] It was found that body-related talk between friends that focused on exercise significantly predicted less body dissatisfaction. [55] Even on SNS, it was indicated that Instagram can provide a positive outlet of self-expression for female through participation in the Body Positive Movement. Instagram content for the '#bodypositive' community was most often positive and inspirational, with a majority of posts promoting body size/image and love. Online communities may serve as sources of social support for young women to discuss mental health concerns and promote physical health, along with a sense of identity and female empowerment. [56] Some detailed examples were included as interventions in later sections.

Regarding to the effect of social and cultural factors on psychology and cognition, it needed to be mentioned that the 'ideal' of body image was perceived differently across culture [57], because we could find differences in expression of weight concerns in different culture. [41], [58] However generally, Asian women had significantly higher scores on the Beliefs About Appearance Scale and the Eating Disorders

Inventory subscales for bulimia, drive for thinness and body dissatisfaction. [59], [60]

Secondly, direct connection between social and cultural environment and weight loss related behaviors were found. [61] Value orientations were shown to be a relevant factor of the use of beauty practices. [62] This relationship could be considered as consequences of change in psychological and cognitive status according to our causal model. Specifically, watching soaps and music television [33], watching TV commercials [31] using thin models, playing with thin dolls [63] during adolescence relates to voluntarily restrained eating or unconscious reduction in food intake. As for peer influence, females who were exposed to a confederate who ate much consumed more than those who were confronted with a confederate who ate only a little or nothing at all. [64] Cultural varieties also exist. Perceptions of overweight and attempts to lose weight were highest in the group of Asian countries where body weights are generally low [2]. In other culture, religious activity could also promote weight loss behaviors, like Ramadan fasting. [65] In addition, increasing monetary cost of dietary energy was associated with both favorable and unfavorable dietary intake patterns and a quite small decrease in BMI in young Japanese women. [66] This is the only evidence found that social factors lead to underweight but do not rely on psychological and cognitive changes.

Thirdly, negative social and cultural environment leads to eating disorders, which matches and support our causal model. Intrasexual competition [67], general SNS use [52] and weight teasing [68] may contribute to the development of disturbed eating and eating disorders in young women.

2.3 Family level

Family relationships are also recognized as one of the contributing factors of underweight. According to the logic web in Figure 2-2, family factors mainly cause underweight through psychological and cognitive change, which then trigger eating disorder that lead to underweight. Though the consequence of eating disorder could be directly induced without passing through weight loss behaviors, it is still hypothesized

that in many cases the development of eating disorder still went through the change of psychological change as a prerequisite.

The most important and fundamental link within familial factors was that negative family relationships leads to negative personality, psychological and cognitive status. Negative parent-adolescent relationships, negative familial communications are linked to negative body image. [43], [69] Particularly, fathers' conversation orientation was negatively and directly related to females' body dissatisfaction. [70] Weight and shape-related positive comments made by siblings were significantly and negatively related to social comparison, which in turn was associated with women's dissatisfaction with their body image. Negative comments were directly linked to body dissatisfaction. [71] In some circumstances, familial factor could also be beneficial. Mothers' conversation orientation was positively associated with females' social competence, which, in turn, was negatively related to their body dissatisfaction. [70] Fathers' endorsement of benevolently sexist beliefs was positively correlated with daughters' weight-related and physical condition body esteem [72].

Family factors could lead to eating disorder directly as found in some studies. It was believed that family dysfunction and parental psychological problems increased the risk of eating disorders in young women. These terms include family difficulties such as problem solving, roles, affective responsiveness, and general functioning, and mother’s somatization, interpersonal sensitivity, depression, anxiety, anger-hostility, paranoid ideation, as well as fathers depression and anger-hostility. [73] Some of these factors were supported in other studies, which include parental overprotectiveness during adolescence [74], domestic violence [75], having been a victim of childhood abuse [76], family histories of depression or eating problems, and changes in family financial circumstances [77]. Living away from parents and poor social adjustment can also increase risk of eating disorder. [78]

In addition to the above dominant causal link, one study was found to relate negative parent-adolescent relationships to dieting behaviors. [69] There was also only one study found showing association between family factors (intimate partner violence) and underweight (malnutrition) directly. [79]

2.4 Individual level

2.4.1 Psychological factors

Psychological factors consist of factors relate to personality, psychological and cognitive status. We classified these three factors together because we thought they share similar properties and have similar effect or are common results of other factors of lower or higher hierarchy.

Firstly, unlike other levels of factors, psychological factors may interact with each other within this level. Because of this, we added an arrow in the causal model pointing to psychological factor itself. Some examples specifically, materialism [19], high appearance importance [41], [80], higher appearance schemas and lower autonomy [81], appearance comparison [82], [83], depression scores [84], [85], negative stress coping [86], perfectionism [87], social physique anxiety [88] are related to a negative body image and risk for developing an eating disorder. Moreover, the concerns with shape were significantly elevated in the group placing a high personal value on shape or weight compare to the group placing a low value on shape and weight, when a depressed mood was induced. [89] In addition to perfectionism, ineffectiveness (feelings of inadequacy, insecurity, worthlessness and having no control over one's life) mediated the relationship between higher baseline self-critical evaluative concerns and eating disorder risk. [87] It was discussed in a study that the underlying reason why young women pursue thinness is that they believe thinness will make them popular [90], and they perceived this as a success to social relationships. [91]

Some contradictory findings exist in the effect of personality. In one study, vulnerable subtypes of narcissism is positively correlated with body dissatisfaction and negatively correlated with self-esteem. [92] While in another study, narcissism was a protective factor, and they discovered that Machiavellianism was a risk factor in the development of body image concerns through self-objectification. [93]

For the effect of religious belief, women with strong spiritual and religious are more likely to pray, meditate, or read religious/spiritual texts to cope with body image distress, which they found helpful. [94] Among a sample of Orthodox Jewish women,

a greater level of extrinsic religious orientation (i.e., religious commitment motivated primarily by societal pressures or expectations) was found predictive of greater disordered eating and body dissatisfaction. A greater level of intrinsic religious orientation (i.e., religious commitment largely motivated from within) was predictive of lower levels of disordered eating and body dissatisfaction. [95]

Secondly, psychological changes directly cause behavioral changes. We observed a significant pattern that negative body image lead to weight-loss behaviors [71], and positive body image reduce weight-loss behaviors. [96] Perceived ideal body attainability was directly and positively related to low-risk and moderate-risk weight control behaviors. [82] Body size dissatisfaction or wish to be thinner was significantly associated with more healthful diets. [97] After adjusting for confounders, underweight subjects with a desire for thinness consumed less cereal and rice, compared to those without a desire for thinness. [98]

Besides body dissatisfaction, we also observed that some psychological factors causing negative body image are also associated with weight-loss behaviors. These include materialism [19], lower body appearance and self-esteem [99]. Also, psychological and social distress drive normal weight woman to embark on a diet and to look for nutritional support in weight loss clinics. [100] After controlling for BMI and trait-based perfectionism, effortless perfectionism uniquely predicted weight control behaviors. [101]

A good explanation for this phenomenon is that women are conflating the "healthy" body and the "ideal" body. [102] Weight misperception seems to be prevalent and is common in normal weight women trying to lose weight. [103] Overestimation of weight [85], distorted perceptions of their fatness [104], body weight status misperception, is associated with weight loss behaviors, resulting in low nutrient intake [105].

Thirdly, as the final consequences, a lot of studies also proved the associations between negative psychological status and eating disorders and underweight. Generally, poor self-reported global and psychological health are associated with underweight. [54] Body image dissatisfaction [106]–[109], widely expressed concern over body image

[110], elevated appearance-ideal internalization [107], [108], weight overestimation [111], body surveillance [108], high personal standard [90] all directly predict eating disorders or related symptoms. Not only body image related psychological status, negative affectivity also highly correlate with eating disorders. [107] This also include low self-esteem [106], [108], negative self-evaluation [90], low self-love or acceptance and high self-blame [112], self-objectification (thinking about and monitoring the body's appearance from an external observer's perspective) [107], extent to which women believed they possessed negative feminine characteristics [113], depression or low mood [114]–[117], perceived stress [115], [118] and anxiety [115] (attachment anxiety, social interaction anxiety [108], anxiety-depression at age 9 [77]). There are also some interactions between the above stated factors. The relation of body image dissatisfaction and eating disorder is buffered by self-esteem, but intensified by social interaction anxiety, body surveillance, internalization of media ideals and attachment anxiety. [108]

Some examples that support the above relationships and needed to be particularly emphasized and summarized. It is generally believed that difficulties with emotion regulation is considered an important maintaining factor of disordered eating. [119], [120] In a study, DE attitudes group had lower levels of Emotional Intelligence (emotional self-awareness, empathy, interpersonal relationships, stress management and happiness) in comparison to the control group. [121] Also, ED patients have significantly lesser average scores in mindfulness, rumination, and coping skills. [122] For example, Instagram users who value audience responses to their selfies are more likely to exhibit disordered eating attitudes and intentions. [123] A high level of guilt when exercise is missed was associated with markedly elevated levels of eating disorder psychopathology and significantly reduced quality of life. [124] Moreover, women with external locus of control (belief that external world has control over the outcome of events in their lives) had significantly greater eating disorder symptomatology than those with an internal locus of control (belief that people themselves have control over the outcome of events in their lives). [125] Elevated levels of Reflective Function (an ability to think about one's own thoughts and feelings) and attenuated alexithymia

(inability to identify and describe emotions experienced by one's self or others), along with elevated cognitive reappraisal and attenuated emotional suppression, were correlated with attenuated ED symptoms. [126] These specific characteristics observed in ED patients help us identify high risk population and allow us to design and deliver intervention precisely.

2.4.2 Behavioral factors

Behavioral factors could be dichotomized to weight-loss behaviors and non-weight-loss behaviors. For weight-loss behaviors, which is the majority of the behavioral factors, they could either cause eating disorder or underweight. Dieting (hypocaloric diets [76], use of diet pills [102], laxatives [127] or detoxes [102], appearance-focused and controlled eating regulation) predicted onset and maintenance of eating disorders [107] or underweight directly [102]. In contrast, autonomous and health-focused eating regulation were associated positively with healthy eating behaviors and were either related negatively or unrelated to disordered eating symptoms. [128] A moderate amount of physical exercise is beneficial to the health, but compulsive exercise, which is characterized by a craving for physical training, resulting in uncontrollable excessive exercise behavior with harmful consequences, such as injuries and impaired social relations, may result in increased frequency of eating-disordered behaviors or underweight. [102], [129], [130]

For non-weight-loss behaviors, overtime work [54] and eating three meals irregularly [131] were found to induce underweight.

2.4.3 Sociodemographic factors

Several sociodemographic factors are associated with elevated level of body dissatisfaction. For age, being younger predicts higher drive for thinness and body dissatisfaction. [91], [132] In terms of race, minority group had higher eating disturbance scores. [133] And as mentioned before, Asian women had significantly higher drive for thinness and body dissatisfaction. [58], [59] As for sexual orientation, heterosexual and bisexual women were more likely to engage in unhealthy weight

control practices overall. [134]

A significant impact of education and wealth on body image, values and physical practices was proven in a study in Slovenia, Croatia and Serbia. [62] This study found that increasing education level of parents (father) are associated with higher frequency of the use of physical beauty practices. Interestingly, estimated value of family assets, monthly income, education of parents and urbanization level of living areas have expectedly had a major impact on shaping the modern (post-traditional) values. However, these factors were found to be positively associated with body image, which is the exactly opposite direction that the authors hypothesized.

2.4.4 Physiological factors

Gene could definitely moderate BMI directly through expression of hormones. [135] Physiological characteristics, such as BMI [99], Body fat [104], and skeletal structure [136] could influence dieting behaviors. These associations are supported by evidence showing BMI [16], [62], [81], [84], [137]–[139] or skeletal structure [136] are associated with weight concerns.

There are also individual varieties in terms of the risk of getting eating disorders. Having type 1 diabetes [140]–[142], ever being overweight [76], and having had serious health problems before age 5 [77] are factors that lead to higher risk of developing eating disorder.

3. Interventions to Prevent or Mitigate Underweight

3.1 Overview

12 studies were recognized as intervention studies. Although strictly speaking, a cross-sectional study (1-6) and a qualitative report (1-11) should not be included, but we thought these also provide significant insight to current project, so we included them as intervention studies.

Interventions can also be classified into the three levels, according to the

ecological model. There are a good number of interventions targeting individual level, and we found one related study that could refer to social and cultural level (1-11), but however, we did not find any interventions targeting family level. Following the model, individual level interventions could be further classified to psychological, behavioral interventions (sociodemographic and physiological factors belong to individual properties and are not changeable).

The majority of the individual level interventions were psychological interventions, these include providing body positive content to improve body image (1-1,1-7,1-8), dissonance-based intervention (1-2,1-9,1-12), mindfulness training (1-3) and meditation training (1-10), and nutrition education (1-5). Behavioral interventions include physical exercise like aerobic training (1-4), and healthy food (cereal) consumption (1-6).

Though the intervention effect varies across or even within studies, no intervention in our searching result was found to be ineffective. There were four studies (1-2, 1-4, 1-9, 1-12) that had more than one intervention studied. Effectiveness of two or more intervention arms was compared, but there was a major one that was of higher interest.

Characteristics of participants vary in a wide range. Included interventions cover a good range of race and ethnicity. There were no particular inclusion criteria for most of the intervention studies, only three (1-4, 1-10, 1-12) mentioned they only recruited young women who were currently having body image concern and wanted to improve body esteem.

3.2 List of successful intervention

1-1 #BoPo on Instagram: An experimental investigation of the effects of viewing body positive content on young women's mood and body image			
Summary	Viewing body positive content on Instagram improves the mood and body image of young women.		
Country	Australia	Year	2019.07
Goal	To investigate the effect of viewing body-positive Instagram posts on young women's mood and body image.		
Study Design	RCT		
Participants	195 women aged 18–30 years old (M = 21.69, SD = 3.49). 52.8% identified		

	as Caucasian, 34.9% Asian (including South East Asian), 5.6% Middle Eastern, 1% Aboriginal or Torres Strait Islander, 0.5% African, and 5.1% identified as 'other' ethnicities. Mean self-reported body mass index (BMI) was 23.08 (SD = 3.90).
Background	Media play a dominant role in influencing perceived social norms and cultural appearance standards, particularly that of the ideal slim female body, commonly referred to as the 'thin-ideal'. A new trend on social media, 'body positivity' aims to challenge these narrow societal prescriptions for female beauty in favour of a broader conceptualization of beauty, body acceptance of all shapes and sizes, and body appreciation.
Methods	● Participants were randomly assigned to one of three exposure conditions (body positive, thin-ideal, or appearance-neutral posts). ● Body positive accounts: (1) @bodyposipanda: images of a larger woman displaying her body with captions about body acceptance, (2) @omgkenzieee: side-by-side images of a 'real' woman challenging societal beauty ideals, (3) @beautyredefined: body positive quotes, and (4) @nolatrees: images of a 'fat' woman practicing yoga with captions focusing on appreciating what her body can do.
Results	● Brief exposure to body positive posts was associated with improvements in young women's positive mood, body satisfaction and body appreciation, relative to thin-ideal and appearance-neutral posts. ● Participants showed favorable attitudes towards the body positive accounts with the majority being willing to follow them in the future.
Conclusions	Body-positive content may offer a fruitful avenue for improving young women's body image.
Link to the Fukuoka Project	This study demonstrated that local government could support body positive content posting accounts so that they can act as a guidance for young women. A different voice in the trend of thin-ideal need to be encouraged by stronger force, and this will benefit young women with better mood and less body dissatisfaction.
Reference	R. Cohen, J. Fardouly, T. Newton-John, and A. Slater, "#BoPo on Instagram: An experimental investigation of the effects of viewing body positive content on young women's mood and body image," NEW MEDIA Soc., vol. 21, no. 7, pp. 1546–1564, Jul. 2019. [143]

1-2 High- and Low-Level Dissonance-Based Eating Disorder Prevention Programs With Young Women With Body Image Concerns: An Experimental Trial			
Summary	Dissonance-based intervention helps to reduce body dissatisfaction and eating disorder related symptoms.		
Country	USA	Year	2011.02
Goal	To test the hypothesis that participants would show greater decreases in thin-ideal internalization, body dissatisfaction, dieting, negative affect, and eating disorder symptoms when given higher level dissonance intervention.		

Study Design	RCT
Participants	124 female undergraduates ranging in age from 18 to 50 years (M = 20.9 years, SD = 3.9). The sample was 8% Asian, 2% Black, 6% Hispanic, 76% White, and 8% specifying other or mixed racial heritage.
Background	Cognitive dissonance is theorized to contribute to the effects of dissonance-based eating disorder prevention programs. The intervention effect needed to be evaluated through experiment.
Methods	● Participants randomly assigned to high- or low-level dissonance intervention group or a control group. ● Dissonance interventions consisted of four sessions with activities including: role-plays wherein participants challenged thin-ideal statements made by facilitators, role-plays wherein participants made quick comebacks to brief thin-ideal statements made by facilitators, discussion of motivations for signing up for the body acceptance group, discussion of possible responses to fat-talk statements (subtle statements that perpetuate the thin ideal), brainstorming about future pressures to be thin and how to avoid or respond.
Results	● Relative to controls, those in the high-dissonance condition showed significantly greater reductions in thin-ideal internalization, body dissatisfaction, dieting, and eating disorder symptoms by posttest. ● Those in the low-dissonance condition showed significantly greater reductions in the first 3 outcomes by posttest, with most of these effects persisting to 3-month follow-up. ● High-dissonance participants showed significantly greater reductions in eating disorder symptoms than low-dissonance participants did by posttest, but this effect was nonsignificant by 3-month follow-up.
Conclusions	Results suggest that dissonance induction contributes to intervention effects but imply that the intervention content, nonspecific factors, and demand characteristics play a much more potent role in producing effects.
Link to the Fukuoka Project	Since the intervention was designed off-line, it might be a good idea to incorporate the intervention content to middle school or high school physical exercise or relevant courses.
Reference	W. McMillan, E. Stice, and P. Rohde, “High- and Low-Level Dissonance-Based Eating Disorder Prevention Programs With Young Women With Body Image Concerns: An Experimental Trial,” J. Consult. Clin. Psychol., vol. 79, no. 1, pp. 129–134, Feb. 2011. [144]

1-3 Does mindfulness have potential in eating disorders prevention? A preliminary controlled trial with young adult women			
Summary	Mindfulness-based intervention is significantly helpful in reducing weight and shape concern, dietary restraint, eating disorder symptoms and related psychosocial impairment.		
Country	Australia	Year	2014

Goal	To assess the feasibility of a pilot mindfulness-based intervention with respect to reducing the risk of eating disorders in young women.
Study Design	RCT
Participants	44 women with body image concerns, aged 17–31 (M = 20.57, SD = 3.22), with a self-reported body mass index (BMI) ranging from 17.72 to 40.00 (M = 24.83, SD = 5.58). The majority were Caucasian (93%), with three participants (7%) identifying as Asian.
Background	Building on the finding that promotion of body acceptance is useful in prevention programmes, a conceivable extension is that acceptance at a higher order level of thoughts, feelings and perceptions may also prove beneficial. Such metacognitive acceptance finds its conceptual roots in mindfulness, which involves the practice of non-judgmental awareness and acceptance of experience in the present moment. With reference to the dual-pathway model, mindfulness may succeed in reducing risk for disordered eating by, first, enabling an increased capacity to refrain from automatic responses when confronted with the thin ideal and related sociocultural pressures, and thereby reducing the ensuing experience and impact of body dissatisfaction, and, second, by reducing the intensity and impact of experiences with a negative affective component (e.g. dissatisfaction, depression) if and when they do occur.
Methods	● Participants were randomly allocated to a mindfulness-based or a dissonance-based intervention (3*1h weekly sessions), or to assessment-only control. ● Self-report measures of eating disorder risk factors, symptoms and related psychosocial impairment were compared at baseline, post-intervention, and at 1- and 6-month follow up. ● Some examples of the intervention contents include thought suppression and magnification exercises, practice of 3-min breathing space, Raisin exercise for maintaining present-moment awareness, reading 'The Guest House' poem.
Results	● At post-intervention, acceptability ratings for both interventions were high. ● Mindfulness participants demonstrated statistically significant improvements relative to control at post-intervention for weight and shape concern, dietary restraint, thin ideal internalization, eating disorder symptoms and related psychosocial impairment; however, these gains were largely lost over follow up. ● Dissonance participants did not show statistically significant improvements relative to control on any outcomes, despite small to moderate effect sizes.
Conclusions	These preliminary findings demonstrate the acceptability and short-term efficacy of a mindfulness-based approach to reducing the risk of disordered eating in young women. This provides support for the continued evaluation of mindfulness in the prevention and early intervention of eating disorders,

	with increased efforts to produce maintenance of intervention gains.
Link to the Fukuoka Project	The intervention content was very comprehensive and well designed, and since the area was closer to psychology, it is strongly recommended to add these contents to psychology class or girl's physiology class to act as a preventive intervention.
Reference	M. J. Atkinson and T. D. Wade, "Does mindfulness have potential in eating disorders prevention? A preliminary controlled trial with young adult women," Early Interv. Psychiatry, vol. 10, no. 3, pp. 234–245, Jun. 2016. [145]

1-4 The effects of aerobic- versus strength-training on body image among young women with pre-existing body image concerns			
Summary	Aerobic-training has higher effect on improving body image than strength-training, which might be helpful in exercise prescription.		
Country	Canada	Year	2014.06
Goal	● To compare the effects of an 8-week supervised program of aerobic exercise versus strength training on changes in body image among young women. ● To examine the relationship between changes in body image and changes in variables that might account for the effects of exercise on body image.		
Study Design	RCT (control group taken as strength-training)		
Participants	Participants were recruited from a university campus through advertisements targeting women who felt dissatisfied with their bodies and who wanted help following through with their New Year's resolutions to exercise. 46 were eligible and enrolled, with mean age 21.5, mean BMI was 22.96, 80% of them were Caucasian. Due to drop-out, finally 23 were in strength-training (ST) condition, and 17 were in (aerobic-training) AT condition.		
Background	Considerable research, including three meta-analyses (Campbell & Hausenblas, 2009; Hausenblas & Fallon, 2006; Reel et al., 2007), indicates that exercise-training programs are effective for improving women's body image. These meta-analyses, however, have produced conflicting results regarding the type of exercise – strength training (e.g., lifting weights, calisthenics) or aerobic training (e.g., walking, cycling) – that is most effective.		
Methods	● Participants were prescribed exercise 3 days/week for 8 weeks, with the goal of completing 24 exercise sessions. Body image, fitness, perceived fitness and exercise self-efficacy were measured before and after the exercise interventions. ● AT condition: exercise was performed on an elliptical exercise machine or a treadmill. Training consisted of a 5 min warm up, 30 min of activity at the target heart rate zone, and a 10 min cool down.		

	● ST condition: training consisted of a very light intensity 5 min warm-up on a bike or treadmill, approximately 30 min of strength training and a 10 min cool down on a bike or a treadmill.
Results	● Both interventions have significant effects on body image measures (appearance evaluation, body areas satisfaction, social physique anxiety), as well as variables for fitness (BMI did not change), perceived fitness and self-efficacy compared to pre-intervention. ● AT had significantly higher effect size on improving appearance evaluation, body areas satisfaction, and reducing social physique anxiety. Most of the effect size of other variables for fitness, perceived fitness and self-efficacy were also higher in AT groups.
Conclusions	● Women in the AT condition reported greater reductions in social physique anxiety (p=.001) and tended to report greater improvements in appearance evaluation (p=.06) than women in the ST condition. ● Changes in perceived fatness, perceived aerobic endurance and aerobic self-efficacy were significantly correlated with body image change (ps<.003).
Link to the Fukuoka Project	Though excessive exercise might induce underweight as described previously, it seems that moderate level of exercise will benefit young women both psychologically and physically. This is extremely helpful since the participants were of normal BMI, which suggest that exercise prescription should be promoted in order to replace unhealthy and risky weight loss behaviors.
Reference	K. A. M. Ginis, H. A. Strong, S. M. Arent, S. R. Bray, and R. L. Bassett-Gunter, “The effects of aerobic- versus strength-training on body image among young women with pre-existing body image concerns,” Body Image, vol. 11, no. 3, pp. 219–227, Jun. 2014. [146]

1-5 Combining Home Garden, Poultry, and Nutrition Education Program Targeted to Families With Young Children Improved Anemia Among Children and Anemia and Underweight Among Nonpregnant Women in Nepal			
Summary	The activity and education-based intervention was tested successful in reducing malnutrition induced anemia and underweight.		
Country	Nepal	Year	2016
Goal	To test the impact of an enhanced homestead food production (EHFP) program, consisting of home garden, poultry raising, and nutrition behavior change communication, on the prevalence of stunting, underweight, wasting, and anemia among young children and underweight and anemia among women.		
Study Design	Cluster RCT		
Participants	2106 and 2614 mother–child pairs at baseline and follow-up from 21 and 30 villages randomized to treatment and control respectively. The location was in Baitadi District, a remote hilly community in the far western region		

	of Nepal.
Background	Nepal has a high prevalence of anemia and underweight. The impact of food-based interventions on child and maternal anthropometry and anemia has not been adequately studied.
Methods	● 1 woman per group of intervention villages was selected and trained by Helen Keller International and they then trained other 20 women on how to establish their home gardens and poultry production (seeds were provided). ● Women met monthly at the VMF to refresh lessons on agriculture techniques and nutrition through social and behavior change communications. ● Project was jointly monitored by officers monthly to follow up and provide additional advice to indicidual family. ● Changes in outcome variables (stunting, underweight, wasting, and anemia among children and underweight and anemia among mothers) were compared between the study groups using mixed-effects logistic regression models.
Results	At follow-up, anemia was significantly lower among children (OR [95% CI]: 0.76 [0.59-0.98]) and mothers (OR [95% CI]: 0.62 [0.48-0.82]) in the treatment group compared to the control. Underweight was lower among mothers in the treatment group compared to the control (OR [95% CI]: 0.61 [0.46-0.82]). There was no impact on child anthropometry.
Conclusions	The EHFP intervention improved anemia among children aged 12 to 48 months and their mothers in Baitadi District of Nepal. The intervention also reduced underweight among these women, but had no impact on child growth, in this district.
Link to the Fukuoka Project	Though the state of development of Japan is different from Nepal, malnutrition due to lack of knowledge or information might still exists. Cooking lessons, nutrition courses could be added to community activities.
Reference	A. Osei et al., “Combining Home Garden, Poultry, and Nutrition Education Program Targeted to Families With Young Children Improved Anemia Among Children and Anemia and Underweight Among Nonpregnant Women in Nepal,” FOOD Nutr. Bull., vol. 38, no. 1, pp. 49–64, Mar. 2017.[147]

1-6 Breakfast Cereal Consumption Moderates the Association Between Body Composition and Body Esteem in Young Women but not in Young Men			
Summary	Usually, women who have higher BMI or body fat tend to have lower body esteem, but this trend was not seen in women who eat cereal during breakfast.		
Country	Canada	Year	2014
Goal	To explore the influence of breakfast cereal consumption on the relationships among BMI, %FM, and body esteem in young adults.		
Study Design	Cross sectional Study		

Participants	28 females (aged 24.6 ± 4.0 years, 15 were cereal consumers) and 29 males (aged 25.1 ± 4.0 years, 15 were cereal consumers) recruited from the Calcium and Dairy Intake Assessment study at the University of Guelph in south-western Ontario.
Background	Body esteem is inversely correlated with Body Mass Index (BMI, kg/m^2) and percent fat mass (%FM). Understanding the determinants of body esteem in young adults is important because of the marked development of health habits during this life stage and the negative consequences of poor body esteem. Consumption of breakfast cereals is also inversely correlated with BMI and %FM in young adults. Therefore, research investigating the relationship among breakfast cereal consumption, body composition, and body esteem is warranted.
Methods	● Participants were identified as breakfast cereal consumers or non-consumers based on their 3-day food record. ● Physical activity, body esteem, BMI and %FM were assessed.
Results	● The %FM was significantly ($P \leq 0.05$) inversely correlated with weight esteem ($r = -0.769$), appearance esteem ($r = -0.723$), and external attribution ($r = -0.620$) in female noncereal consumers. ● BMI was similarly correlated with body esteem scores. ● These relationships were not significant in female cereal consumers (all $r < 0.426$), despite no difference in confounding variables between female cereal consumers and non-consumers.
Conclusions	Breakfast cereal consumption may moderate the relationship among BMI, %FM, and body esteem in young adult women and may be useful for improving body esteem without focusing on weight loss.
Link to the Fukuoka Project	This was not an intervention study but could give us insight on how the food we eat modifies our body image. Healthy food not only benefit our health physically, but also make us feel good psychologically from the perspective of body esteem. Dining halls of schools or workplace could introduce cereal in breakfast or other so-called healthy food into their menus.
Reference	H.-A. Scott, M. Marcinow, R. Hicks, and A. C. Buchholz, “Breakfast Cereal Consumption Moderates the Association Between Body Composition and Body Esteem in Young Women but not in Young Men,” Can. J. Diet. Pract. Res., vol. 75, no. 4, pp. 214–217, Dec. 2014.[148]

1-7 The effect of exposure to parodies of thin-ideal images on young women’s body image and mood			
Summary	Women who view Instagram post that challenge the thin-ideal images had improvements in their body image and mood.		
Country	UK	Year	2019.06
Goal	To experimentally investigate the influence of exposure to humorous, parody images of thin-ideal celebrity Instagram posts on women’s body		

	satisfaction and mood compared to exposure to thin-ideal celebrity posts alone.
Study Design	RCT
Participants	102 women aged 18–30 years old (M = 23.55, SD = 2.33), recruited through social media via convenience sampling. They had a mean body mass index of 22.97 (SD = 3.91). The majority of participants identified as White (n = 95, 93.1%), 5 identified as mixed race (4.9%), and 2 identified as Asian (2.0%).
Background	● Although social networking services typically promote the thin beauty ideal for women, they also provide an opportunity for users to challenge this dominant ideal in unique and novel ways. ● A popular public figure on Instagram is Celeste Barber, Celeste is an average-sized woman, and her posts underscore the absurdity of popular celebrity posts by highlighting the unrealistic and generally unattainable nature of such images. In this way, they might be considered a form of media literacy, which aims to enhance critical thinking and skepticism about media in an attempt to reduce its persuasive influence. ● There is much research demonstrating the broader psychological benefits of humor. While there is no research to date that has examined the impact of humorous stimuli on body image.
Methods	● Two sets of visual stimuli were created for the study, each containing 16 images. The parody images were sourced from Celeste Barber's Instagram account. The thin-ideal images were sourced from celebrities' public Instagram accounts. The parody set contained images of Celeste Barber's recreations of celebrity Instagram posts, paired with the original photo. The thin-ideal set contained only the images of the celebrities used in Celeste Baber's posts. ● Data on Instagram usage, state body satisfaction and mood, state body appreciation, state appearance comparisons, social media literacy, trait internalization o the thin ideal were collected before and after the exposure.
Results	● Results indicated that acute exposure to parody images led to increased body satisfaction and positive mood (happiness) compared to exposure to the thin-ideal celebrity images alone. ● No group differences were found on levels of trait appearance comparison or social media literacy, and the findings were not moderated by trait levels of thin-ideal internalization.
Conclusions	The findings of the current study provide preliminary evidence that parody images of thin-ideal celebrity Instagram posts may provide a novel and useful approach for increasing body satisfaction and positive mood in young women.
Link to the Fukuoka Project	The content of these post or images could be used as education materials, and could be posted electronically through school, company websites or

	topics of social network services. More account and content with critical view and innovative thoughts need to be supported.
Reference	A. Slater, N. Cole, and J. Fardouly, “The effect of exposure to parodies of thin-ideal images on young women’s body image and mood,” Body Image, vol. 29, pp. 82–89, Jun. 2019.[149]

1-8 Healthy Body Image Intervention Delivered to Young Women via Facebook Groups: Formative Study of Engagement and Acceptability			
Summary	A Facebook group-based intervention with high acceptance and engagement was helpful in improving body image, which is reflected by their frequency of indoor tanning.		
Country	USA	Year	2018.02
Goal	To evaluate engagement with and acceptability of using a secret Facebook group to deliver a healthy body image intervention to young women engaged in indoor tanning.		
Study Design	Pre-post intervention study		
Participants	Participants were 18 to 25-year-old women who used Facebook at least once a day and a tanning bed at least once in the past year. A total of 17 participants were enrolled (6 recruited from prior focus groups, 3 campus flyers, 6 social media and Craigslist, and 2 referrals from enrolled participants).		
Background	There is increasing interest in using social media sites such as Facebook to deliver health interventions so as to expose people to content while they are engaging in their usual social media habit. This formative intervention development study is novel in describing a preliminary test of using the secret group feature of Facebook to deliver a behavioral intervention targeting users of indoor tanning beds to reduce their risk of skin cancer.		
Methods	● Participants completed a baseline survey and joined a secret Facebook group with intervention content delivered via daily posts for 4 weeks. ● Some contents posted in the group were: 1) Information-based content intended to provide context for the intervention content and counter-perspectives to idealistic thinking. Designed to raise participants’ awareness of their thoughts, feelings, and actions with regard to their appearance. 2) Inspirational or humorous quotes related to resisting idealistic thinking, empowerment, or body activism. 3) Questions or discussion topics that elicited comments and responses from group members. 4) Skill-building, homework-type activities were programmed with online survey software. ● Engagement data was extracted, and acceptability was measured via a follow-up survey, and number of past 1-month tanning sessions was measured on the baseline and follow-up assessment		
Results	● On average, posts were viewed by 91% of participants, liked by 35%, and commented on by 26%. The average comment rate was highest		

	(65%) for posts that elicited comments by directly posing questions or discussion topics to the group. ● Average intervention acceptability ratings were highly positive, and participants reported feeling connected to the group and its topic. ● Average rates of past 1-month indoor tanning reported following the intervention were lower than the baseline rate (P=.08, Cohen d=0.47).
Conclusions	Social media represents an unexplored platform for delivering dissonance-based interventions for disordered eating and indoor tanning. The current formative study demonstrates the feasibility and acceptability of using Facebook as a delivery mechanism for these types of interventions. The use of Facebook groups to deliver dissonance-based interventions capitalizes on features key to the success of these interventions including group-based discussions and peer support.
Link to the Fukuoka Project	This study demonstrated the feasibility of this online intervention with low cost and high engagement aiming to reduce eating disorders and body dissatisfaction in a group basis. It also shows potential for creating highly sustainable intervention approaches with strong potential for dissemination across multiple geographic locations.
Reference	J. L. Stapleton, S. L. Manne, A. K. Day, K. Levonyan-Radloff, and S. L. Pagoto, "Healthy Body Image Intervention Delivered to Young Women via Facebook Groups: Formative Study of Engagement and Acceptability," JMIR Res. Protoc., vol. 7, no. 2, Feb. 2018.[150]

1-9 A Preliminary Trial of a Prototype Internet Dissonance-Based Eating Disorder Prevention Program for Young Women With Body Image Concerns			
Summary	Four types of intervention were compared in terms of the efficacy in improving eating disorder related symptoms. It was shown that the order of the efficacy was group intervention (Body project) > Internet intervention (eBody project) >> video control > brochure control.		
Country	USA	Year	2012
Goal	● To test whether the eBody Project would produce significantly larger reductions in eating disorder risk factors and symptoms relative to the educational video and brochure control conditions. ● To test whether the eBody Project produced reductions in outcomes that were similar to the reductions observed for the group Body Project conducted by college counselors, which was also expected to produce larger effects than the educational video and brochure control conditions. ● To benchmark the magnitude of the intervention effects of the eBody Project against effects for the Body Project from our efficacy trial		
Study Design	RCT		
Participants	107 young women (Mean age = 21.6 years, SD = 6.6) recruited from a local university with a mean BMI of 23.7 kg/m^2 (SD = 4.3).		

Background	A group dissonance-based eating disorder prevention program, in which young women critique the thin ideal, reduces eating disorder risk factors and symptoms, but it can be difficult to identify school clinicians with the time and expertise to deliver the intervention.
Methods	● Participants were randomized to the Internet intervention, group intervention, educational video condition, or educational brochure condition. (last two were controls) ● Intervention content: 1) Internet intervention: participants were asked to complete six modules in which they engage in written and behavioral activities designed to critique the thin-ideal over the course of 3 weeks; 2) Group intervention: participants voluntarily engaged in verbal, written, and behavioral exercises in which they critiqued the thin ideal during four 1-hr group sessions held on a weekly basis and in home exercises. ● Thing-ideal internalization, body dissatisfaction, dieting, negative affect and eating pathology were tested before and after interventions.
Results	● Internet and group participants showed greater pre-post reductions in eating disorder risk factors and symptoms than video controls (M ds = 0.47 and 0.54, respectively) and brochure controls (M ds = 0.75 and 0.72, respectively), with many effects reaching significance. ● Effects did not differ significantly for Internet versus group participants (M ds = –0.13) or for video versus brochure controls (M d = 0.25). ● Effect sizes for the Internet intervention were similar to those previously observed for group versions of this intervention.
Conclusions	Results suggest that this prototype Internet intervention is as efficacious as the group intervention, implying that there would be merit in completing this intervention and evaluating it in a fully powered trial.
Link to the Fukuoka Project	This study provides important evidence on the priority of different types of interventions. Although intervention effect usually correlated with its costs, we could still use this information to find the appropriate one with the best cost-benefit ratio during decision making.
Reference	E. Stice, P. Rohde, S. Durant, and H. Shaw, “A Preliminary Trial of a Prototype Internet Dissonance-Based Eating Disorder Prevention Program for Young Women With Body Image Concerns,” J. Consult. Clin. Psychol., vol. 80, no. 5, pp. 907–916, Oct. 2012.[151]

1-10 Brief self-compassion meditation training for body image distress in young adult women			
Summary	Self-compassion training can impact aspects of self-compassion and body image dissatisfaction.		
Country	USA	Year	2016.12
Goal	To evaluate a version of an Internet-based self-compassion training, which had previously shown promising results, but was limited by high attrition.		

Study Design	RCT
Participants	80 undergraduate women with body image/appearance concerns and who were not currently meditating. The women ranged in age from 18 to 21 (M = 18.85, SD = 0.87), with a mean body mass index (BMI) of 22.20 (SD=3.60), based on self-reported height and weight.
Background	● Self-compassion interventions may be uniquely suited to address body image distress (BID), as change-based strategies may have limited utility in a cultural context that so highly values appearance. ● The current study evaluated a version of an Internet-based self-compassion training, which had previously shown promising results, but was limited by high attrition.
Methods	● All training exercises were in the form of 20-minute podcasts. The self-compassion meditation exercises were those used by Albertson et al. (2014). The three exercises included a compassionate body scan (completed first in the lab and then sent to participants the following two days), an affectionate breathing exercise (sent to participants on Days 4 and 5), and a loving-kindness meditation directed toward the body (sent to participants on Days 6 and 7). ● Self-compassion, body appreciation, self-esteem, body surveillance, body shame, appearance-contingent self-worth and body dissatisfaction were assessed before and after the intervention.
Results	Consistent with the prior study, these analyses indicated that intervention participants showed significantly greater pre-post change on body appreciation and appearance-contingent self-worth (as well as body surveillance) compared to controls. Though significantly greater reductions in body shame or body dissatisfaction following training were not observed.
Conclusions	Results suggest that brief exposure to the basic tenets of self-compassion holds promise for improving aspects of self-compassion and BID.
Link to the Fukuoka Project	Similar to dissonance intervention, this is another type of effective method that purely based on psychological improvement. Thus, weekly community activities and school physical class are good place to incorporate this intervention. Since the social cultural environment might be hard to change, being strong from inside could be more effective.
Reference	A. M. Toole and L. W. Craighead, “Brief self-compassion meditation training for body image distress in young adult women,” Body Image, vol. 19, pp. 104–112, Dec. 2016.[152]

1-11 Swimsuit issues: promoting positive body image in young women’s magazines			
Summary	A Code of using models with diverse range of body shapes and sizes in fashion, media and advertising industries was proven successful in Australia.		
Country	Australia	Year	2011.08

Goal	To review the promotion of healthy body image to young Australian women, following the 2009 introduction of the voluntary Industry Code of Conduct on Body Image. The Code includes using diverse sized models in magazines.
Study Design	Qualitative content analysis
Participants	10 Australian young women's magazines
Background	Voluntary Industry Code of Conduct on Body Image was introduced in 2009. The Code was part of a wider campaign following the Australian Government's response to the Body Image Advisory Group founded in 2009, amidst international concern about fashion and body image promotion. One of the Group's key recommendations was the drafting of the Code in order to provide fashion, media and advertising industries with a guideline for the endorsement of healthy body image. The Code advocates the use of a diverse range of body shapes and sizes, and to refrain from any digital manipulation that alters body shapes to unrealistic standards. Research has suggested that the promotion of models with more average or fuller figures can reduce the levels of body dissatisfaction and anxiety experienced by female readers. While the Code is not compulsory, it aims to encourage body-friendly health promotion campaigns and strategies.
Methods	Each magazine was manually coded according to three separate criteria listed below: ● Pictorial evidence of diverse size models – the swimsuit feature(s) included at least one fuller size/plus size model. ● Pictorial/textual evidence of body size differentials – several body sizes/shapes are represented in the swimsuit feature(s). ● Textual evidence of diverse body type enhancement tips – the swimsuit feature(s) included helpful tips/comments directed towards different body types.
Results	Diverse sized models were incorporated in four of the seven magazines with swimsuit features sampled. Body size differentials were presented as part of the swimsuit features in three of the magazines sampled. Tips for diverse body type enhancement were included in four of the magazines. All magazines met at least one criterion. One magazine displayed evidence of all three criteria.
Conclusions	Preliminary examination suggests that more than half of young women's magazines are upholding elements of the voluntary Code of Conduct for Body Image, through representation of diverse-sized women in their swimsuit issues.
Link to the Fukuoka Project	The meaning of this study is to show that how Australia voluntarily formed this Code aiming to promote people's body image satisfaction. Result demonstrated that the Code was implemented successfully, which pioneered and provide precious reference for the enact of such code in Japan. This study is the only related intervention targeting social cultural level in this review.

Reference	E. R. Boyd and J. Moncrieff-Boyd, "Swimsuit issues: promoting positive body image in young women's magazines," Heal. Promot. J. Aust., vol. 22, no. 2, pp. 102–106, Aug. 2011.[153]

1-12 Online prevention of disordered eating in at-risk young-adult women: a two-country pragmatic randomized controlled trial			
Summary	Two different interventions, one focused on behavioral aspects and the other focused on cognitive components, were compared. The one targeting cognitive aspects were proved to have higher ability in preventing disordered eating.		
Country	Australia (and New Zealand)	Year	2018.09
Goal	To evaluate the 'real world' effectiveness of an established online ED prevention program (SB) relative to a new online program (MS-T) and a control condition with young-adult women.		
Study Design	RCT		
Participants	608 women aged 18–25 years (M age = 20.71 years) from Australia and New Zealand who wished to improve their body image.		
Background	● Disordered eating (DE) is a widespread, serious problem. Efficacious prevention programs that can be delivered at-scale are needed. ● To date SB has not been compared with another intervention. In the current study we compared it with Media Smart-Targeted (MS-T), an adapted online version of an 8-lesson school-based program (MS) that has been found to reduce ED risk factors relative to controls and active interventions in young-adolescent (Grades 7 and 8) girls and boys. ● This research extended to a general community population as opposed to university and school-based settings typically investigated; and, allowed participants with higher baseline levels of eating pathology than in previous trials.		
Methods	● Participants were randomized to three groups: Media Smart-Targeted (MS-T) and Student Bodies (SB) were both 9-module interventions released weekly, whilst control participants received positive body image information. ● MS-T and SB: Both programs released modules weekly via a password-protected mobile internet-platform and both programs were a pure self-help format (no online therapist moderator). Participants were encouraged to view content (e.g. written, video), respond to questions, and complete accompanying assignments. SB focused on behavioral targets (e.g. eating, exercise) that are proximal to DE in risk factor models while MS-T focused more on cognitive components (e.g. media internalization, ineffectiveness) that are more distal in such models. ● Primary [Eating Disorder Examination–Questionnaire (EDE-Q) Global], secondary (DE risk factors) and tertiary (DE) outcome		

	measures were completed at baseline, post-program, 6- and 12-month follow-up.
Results	● Primary intent-to-treat (ITT) analyses revealed no differences between groups, while measure completer analyses found MS-T had significantly lower EDE-Q Global than controls at 12-month follow-up. ● Secondary ITT analyses found MS-T participants reported significantly higher quality of life–mental relative to both SB and controls (6-month follow-up), while MS-T and controls had lower clinical impairment relative to SB (post-program). Amongst measure completers, MS-T scored significantly lower than controls and SB on 5 variables. ● Of those with baseline DE, MS-T participants were significantly less likely than controls to have DE at 12-month follow-up.
Conclusions	Given both programs were not therapist-moderated, MS-T has potential to achieve reductions in disordered eating risk at low implementation costs.
Link to the Fukuoka Project	This research also provided extremely meaningful suggestion on the priority of interventions targeting different factors of DE. It is thus recommended that interventions from the cognitive perspective should have higher priority of implementation since they have higher efficacy. Content of these interventions could also be referred to.
Reference	S. M. Wilksch, A. O'Shea, C. B. Taylor, D. Wilfley, C. Jacobi, and T. D. Wade, "Online prevention of disordered eating in at-risk young-adult women: a two-country pragmatic randomized controlled trial," Psychol. Med., vol. 48, no. 12, pp. 2034–2044, Sep. 2018.[154]

4. Findings and conclusions

The major finding of this report is that there is an underlying complex and interacting logic web that explains the onset of underweight. A hierarchical model was established and some major pathways that lead to underweight were identified. Specifically, from the highest level to the lowest in the hierarchy are: (1a) social and cultural environment with thin-ideal as the dominant value, (1b) negative family relationships, (1c) Individual physiological or sociodemographic factors, (2) individual negative psychological or cognitive status, (3) individual weight-loss behaviors, and finally (4) eating disorders or underweight.

As we can see from the above association web, social and cultural influence is the

root in the cause of the problem. It could be imagined that as long as this 'ceilings' exist, underweight could not be eliminated. Human beings are societal animals, we are constantly exchanging information with outside world for better survival both physically and psychologically. Thus, in order to give more room for perceived range of healthy BMI, interventions targeting social and cultural level should be considered as priorities. In some European countries, fashion models are required to have a BMI of 18 kg/m^2 or greater to discourage severe emaciation. And we could see the effect of this kind of social code in intervention 1-11. This calls for the need of enacting such code in media industry to prevent young women from suffering from the gaps between their so-called 'ideals' and reality. Comparing to societal or administrative cost, potential benefit of reduction in medical resource consumption with the reduction of underweight and its complications is considerable.

On the other hand, since it is hardly feasible to carry out intervention at family level or targeting physiological or sociodemographic factors, psychological status becomes the second influential perspective. From previous analysis, psychological reason is also the direct cause of many weight-loss behaviors, onset of eating disorders and underweight. Thus, it is the second priority to implement psychological interventions in young women population. In the list of successful interventions, we found as many as 9 studies (75% of all interventions) are targeting individual psychological or cognitive aspects. Most of them are educational or training programs that could be easily incorporated to physical education course, psychology course, physiological course (biology course) or just after-class activity. As is shown previously that vulnerable people have lower reflective functions, it might be less likely to attend these courses voluntarily. Some administrative forces might be needed to make some of the intervention content implemented as compulsory. Since it is believed that values are formed at a young age, it is recommended that psychological or educational interventions need to be delivered early, for example, sometime around early adolescence when sexual development starts. In addition, although we could not change family history of disease or harmony of family relationships, it is worth trying some interventions that improve parents' knowledge on nutrition, as well as the importance

of parental behaviors and attitudes in shaping their child's behaviors and values. For example, but the form of parent class meeting, or pamphlets passing to them through their children.

Not restricted to present evidence of effective intervention, especially there is only evidence for societal level intervention, it is always beneficial to test something new. Since Japan is one of the major countries with underweight problem, it is of higher significance to design and test more interventions. This will create directly interpretable and implementable evidence in Japan. There is no need for further emphasis on how important the strategic position is. For example, Ministry of Health or local government could foster some channels on SNS broadcasting health related contents, then include above intervention contents as one of the chapters. One possible challenge might be that the effectiveness of social and cultural interventions is hard to test. Nevertheless, basing on present evidence, we should have the confidence to come up with more new ideas and it is always no harm to make efforts for the purpose of the health of residents.

With the solid evidence of the effectiveness of these interventions, and the high motivation to provide more with research investment, we believe that the social and cultural environment will be more diversified in values on beauty, and young women will be better equipped with more knowledge and higher confidence.

References

[1] N. Kiriike, T. Nagata, K. Sirata, and N. Yamamoto, "Are young women in Japan at high risk for eating disorders?: Decreased BMI in young females from 1960 to 1995," *PSYCHIATRY Clin. Neurosci.*, vol. 52, no. 3, pp. 279–281, Jun. 1998.

[2] J. Wardle, A. M. Haase, and A. Steptoe, "Body image and weight control in young adults: International comparisons in university students from 22 countries," *Int. J. Obes.*, vol. 30, no. 4, pp. 644–651, Apr. 2006.

[3] A. Sugawara, K. Saito, M. Sato, S. Kodama, and H. Sone, "Thinness in Japanese Young Women," *EPIDEMIOLOGY*, vol. 20, no. 3, pp. 464–465, May 2009.

[4] Y. Takasaki, T. Fukuda, Y. Watanabe, T. Kurosawa, and K. Shigekawa, "Ideal body shape in young Japanese women and assessment of excessive leanness based on allometry," *J. Physiol. Anthropol. Appl. Human Sci.*, vol. 22, no. 2, pp. 105–110, Mar. 2003.

[5] "Eating disorder - Wikipedia." [Online]. Available: https://en.wikipedia.org/wiki/Eating_disorder#cite_note-DSM-5-10. [Accessed: 03-Mar-2020].

[6] "Health Consequences | National Eating Disorders Association." [Online]. Available: https://www.nationaleatingdisorders.org/health-consequences. [Accessed: 03-Mar-2020].

[7] J. Treasure, A. M. Claudino, and N. Zucker, "Eating disorders," *The Lancet*, vol. 375, no. 9714. Lancet, pp. 583–593, 2010.

[8] J. Lim and H. S. Park, "Relationship between underweight, bone mineral density and skeletal muscle index in premenopausal Korean women," *Int. J. Clin. Pract.*, vol. 70, no. 6, pp. 462–468, Jun. 2016.

[9] A. I. Girsen *et al.*, "Women's prepregnancy underweight as a risk factor for preterm birth: a retrospective study," *BJOG An Int. J. Obstet. Gynaecol.*, vol. 123, no. 12, pp. 2001–2007, Nov. 2016.

[10] L. Roh, J. Braun, A. Chiolero, M. Bopp, S. Rohrmann, and D. Faeh, "Mortality risk associated with underweight: A census-linked cohort of 31,578 individuals with up to 32 years of follow-up," *BMC Public Health*, vol. 14, no. 1, p. 371, Apr. 2014.

[11] "Underweight health risks: Causes, symptoms, and treatment." [Online]. Available: https://www.medicalnewstoday.com/articles/321612#risks. [Accessed: 03-Mar-2020].

[12] M. Siervo *et al.*, "Weight loss expectations and body dissatisfaction in young women attempting to lose weight," *J. Hum. Nutr. Diet.*, vol. 27, no. 2, pp. 84–89, Apr. 2014.

[13] A. L. Ahern, K. M. Bennett, and M. M. Hetherington, "Internalization of the Ultra-Thin Ideal: Positive Implicit Associations with Underweight Fashion Models are Associated with Drive for Thinness in Young Women," *Eat. Disord.*, vol. 16, no. 4, pp. 294–307, Jul. 2008.

[14] D. J. Anschutz, D. Spruijt-Metz, T. Van Strien, and R. C. M. E. Engels, "The direct effect of thin ideal focused adult television on young girls' ideal body figure," *Body Image*, vol. 8, no. 1, pp. 26–33, Jan. 2011.

[15] I. Prichard, A. C. McLachlan, T. Lavis, and M. Tiggemann, "The Impact of Different Forms of #fitspiration Imagery on Body Image, Mood, and Self-Objectification among Young Women," *Sex Roles*, vol. 78, no. 11–12, pp. 789–798, Jun. 2018.

[16] H. Chen, X. Gao, and T. Jackson, "Predictive models for understanding body dissatisfaction among young males and females in China," *Behav. Res. Ther.*, vol. 45, no. 6, pp. 1345–1356, Jun. 2007.

[17] L. Shorter, S. L. Brown, S. J. Quinton, and L. Hinton, “Relationships between body-shape discrepancies with favored celebrities and disordered eating in young women,” *J. Appl. Soc. Psychol.*, vol. 38, no. 5, pp. 1364–1377, May 2008.
[18] T. Jackson, P. Zheng, and H. Chen, “Features of objectified body consciousness and sociocultural perspectives as predictors of eating and body image disturbances among young women and men in China,” *J. Gend. Stud.*, vol. 25, no. 5, pp. 599–612, 2016.
[19] U. Gudnadottir and R. B. Gardarsdottir, “The influence of materialism and ideal body internalization on body-dissatisfaction and body-shaping behaviors of young men and women: Support for the Consumer Culture Impact Model,” *Scand. J. Psychol.*, vol. 55, no. 2, pp. 151–159, Apr. 2014.
[20] I. Nagar and R. Virk, “The Struggle Between the Real and Ideal: Impact of Acute Media Exposure on Body Image of Young Indian Women,” *SAGE OPEN*, vol. 7, no. 1, Jan. 2017.
[21] H. D. Posavac, S. S. Posavac, and E. J. Posavac, “Exposure to media images of female attractiveness and concern with body weight among young women,” *Sex Roles*, vol. 38, no. 3–4, pp. 187–201, Feb. 1998.
[22] T. L. Rochelle and W. Y. Hu, “Media influence on drive for thinness, body satisfaction, and eating attitudes among young women in Hong Kong and China,” *Psychol. Heal. Med.*, vol. 22, no. 3, pp. 310–318, 2017.
[23] L. Yi and L. Dongbiao, “The Influence of Media on Young Chinese Female’s Body Image,” in *PROCEEDINGS OF 2011 INTERNATIONAL SYMPOSIUM - THE FEMALE SURVIVAL AND DEVELOPMENT*, 2011, pp. 108–112.
[24] U.-J. Yu and J. Jung, “Effects of Self-Discrepancy and Self-Schema on Young Women’s Body Image and Self-Esteem after Media Image Exposure,” *Fam. Consum. Sci. Res. J.*, vol. 47, no. 2, pp. 142–160, Dec. 2018.
[25] N. L. Wood-Barcalow, T. L. Tylka, and C. L. Augustus-Horvath, “``But I Like My Body{''}: Positive body image characteristics and a holistic model for young-adult women,” *Body Image*, vol. 7, no. 2, pp. 106–116, Mar. 2010.
[26] T. Jackson, C. Jiang, and H. Chen, “Associations between Chinese/Asian versus Western mass media influences and body image disturbances of young Chinese women,” *Body Image*, vol. 17, pp. 175–183, Jun. 2016.
[27] D. Schooler, L. M. Ward, A. Merriwether, and A. Caruthers, “Who’s that girl: Television’s role in the body image development of young white and black women,” *Psychol. Women Q.*, vol. 28, no. 1, pp. 38–47, Mar. 2004.
[28] A. Carter, J. I. Forrest, and A. Kaida, “Association Between Internet Use and Body Dissatisfaction Among Young Females: Cross-Sectional Analysis of the Canadian Community Health Survey,” *J. Med. INTERNET Res.*, vol. 19, no. 2, Feb. 2017.
[29] A. Becker, “Television, disordered eating, and young women in Fiji: Negotiating body image and identity during rapid social change,” *Cult. Med. PSYCHIATRY*, vol. 28, no. 4, pp. 533–559, Dec. 2004.
[30] H. Dohnt and M. Tiggemann, “The contribution of peer and media influences to the development of body satisfaction and self-esteem in young girls: A prospective study,” *Dev. Psychol.*, vol. 42, no. 5, pp. 929–936, Sep. 2006.
[31] D. J. Anschutz, R. C. M. E. Engels, E. S. Becker, and T. Van Strien, “The effects of TV commercials using less thin models on young women’s mood, body image and actual food intake,” *Body Image*, vol. 6, no. 4, pp. 270–276, Sep. 2009.
[32] P. N. MYERS and F. A. BIOCCA, “THE ELASTIC BODY-IMAGE - THE EFFECT OF

TELEVISION ADVERTISING AND PROGRAMMING ON BODY-IMAGE DISTORTIONS IN YOUNG-WOMEN," *J. Commun.*, vol. 42, no. 3, pp. 108–133, 1992.

[33] D. Anschutz, R. Engels, J. Van Leeuwe, and T. van Strien, "Watching your weight? The relations between watching soaps and music television and body dissatisfaction and restrained eating in young girls," *Psychol. Health*, vol. 24, no. 9, pp. 1035–1050, 2009.

[34] H. K. Dohnt and M. Tiggemann, "Body image concerns in young girls: The role of peers and media prior to adolescence," *J. Youth Adolesc.*, vol. 35, no. 2, pp. 141–151, Apr. 2006.

[35] C. Bell and M. J. Cooper, "Socio-cultural and cognitive predictors of eating disorder symptoms in young girls.," *Eat. Weight Disord.*, vol. 10, no. 4, 2005.

[36] D. Greenwood, "Idealized TV friends and young women's body concerns," *Body Image*, vol. 6, no. 2, pp. 97–104, Mar. 2009.

[37] M. Girard, R. F. Rodgers, and H. Chabrol, "Prospective predictors of body dissatisfaction, drive for thinness, and muscularity concerns among young women in France: A sociocultural model," *Body Image*, vol. 26, pp. 103–110, Sep. 2018.

[38] D. M. Gondoli, A. F. Corning, E. H. B. Salafia, M. M. Bucchianeri, and E. E. Fitzsimmons, "Heterosocial involvement, peer pressure for thinness, and body dissatisfaction among young adolescent girls," *Body Image*, vol. 8, no. 2, pp. 143–148, Mar. 2011.

[39] T. Jackson and H. Chen, "Sociocultural predictors of physical appearance concerns among adolescent girls and young women from China," *Sex Roles*, vol. 58, no. 5–6, pp. 402–411, Mar. 2008.

[40] J. W. Kim, "Facebook Use for Profile Maintenance and Social Grooming and Young Korean Women's Appearance Comparison With Peers and Body Image Concerns," *Soc. MEDIA + Soc.*, vol. 4, no. 2, May 2018.

[41] K. Murray, E. Rieger, and D. Byrne, "The Effect of Peer Stress on Body Dissatisfaction in Female and Male Young Adults," *J. Exp. Psychopathol.*, vol. 7, no. 2, pp. 261–276, 2016.

[42] J. V Hogue and J. S. Mills, "The effects of active social media engagement with peers on body image in young women," *Body Image*, vol. 28, pp. 1–5, Mar. 2019.

[43] J. C. Kichler and J. H. Crowther, "Young Girls' Eating Attitudes and Body Image Dissatisfaction Associations With Communication and Modeling," *J. Early Adolesc.*, vol. 29, no. 2, pp. 212–232, Apr. 2009.

[44] M. Dhillon and P. Dhawan, "``But I am fat{''}: The experiences of weight dissatisfaction in Indian adolescent girls and young women," *WOMENS Stud. Int. FORUM*, vol. 34, no. 6, pp. 539–549, 2011.

[45] C. N. Markey and P. M. Markey, "Romantic relationships and body satisfaction among young women," *J. Youth Adolesc.*, vol. 35, no. 2, pp. 271–279, Apr. 2006.

[46] H. K. Dohnt and M. Tiggemann, "Peer influences on body dissatisfaction and dieting awareness in young girls," *Br. J. Dev. Psychol.*, vol. 23, no. 1, pp. 103–116, Mar. 2005.

[47] M. Stel and G. M. van Koningsbruggen, "Healthy food consumption in young women. The influence of others' eating behavior and body weight appearance," *Appetite*, vol. 90, pp. 240–247, Jul. 2015.

[48] J. Fardouly, P. C. Diedrichs, L. R. Vartanian, and E. Halliwell, "Social comparisons on social media: The impact of Facebook on young women's body image concerns and mood," *Body Image*, vol. 13, pp. 38–45, Mar. 2015.

[49] J. Fardouly, B. K. Willburger, and L. R. Vartanian, "Instagram use and young women's body image concerns and self-objectification: Testing mediational pathways," *NEW MEDIA Soc.*, vol. 20,

no. 4, pp. 1380–1395, Apr. 2018.
[50] R. Cohen, T. Newton-John, and A. Slater, “The relationship between &ITFacebook&IT and &ITInstagram&IT appearance-focused activities and body image concerns in young women,” *Body Image*, vol. 23, pp. 183–187, Dec. 2017.
[51] R. Cohen, T. Newton-John, and A. Slater, “`Selfie’-objectification: The role of selfies in self-objectification and disordered eating in young women,” *Comput. Human Behav.*, vol. 79, pp. 68–74, Feb. 2018.
[52] L. M. Howard, K. E. Heron, R. I. MacIntyre, T. A. Myers, and R. S. Everhart, “Is use of social networking sites associated with young women’s body dissatisfaction and disordered eating? A look at Black-White racial differences,” *Body Image*, vol. 23, pp. 109–113, Dec. 2017.
[53] J. S. Mills, S. Musto, L. Williams, and M. Tiggemann, “``Selfie{’’} harm: Effects on mood and body image in young women,” *Body Image*, vol. 27, pp. 86–92, Dec. 2018.
[54] S. M. Ali and M. Lindstrom, “Socioeconomic, psychosocial, behavioural, and psychological determinants of BMI among young women: differing patterns for underweight and overweight/obesity,” *Eur. J. Public Health*, vol. 16, no. 3, pp. 324–330, Jun. 2006.
[55] L. Wasylkiw and M. E. Williamson, “Actual Reports and Perceptions of Body Image Concerns of Young Women and Their Friends,” *Sex Roles*, vol. 68, no. 3–4, pp. 239–251, Feb. 2013.
[56] L. Kelly and S. Daneshjoo, “INSTAGRAM & BODY POSITIVITY AMONG FEMALE ADOLESCENTS & YOUNG ADULTS,” *Ann. Behav. Med.*, vol. 53, no. 1, p. S599, Mar. 2019.
[57] M. Tiggemann, A. Verri, and S. Scaravaggi, “Body dissatisfaction, disordered eating, fashion magazines, and clothes: A cross-cultural comparison between Australian and Italian young women,” *Int. J. Psychol.*, vol. 40, no. 5, pp. 293–302, Oct. 2005.
[58] N. Mahmud and N. Crittenden, “A comparative study of body image of Australian and Pakistani young females,” *Br. J. Psychol.*, vol. 98, no. 2, pp. 187–197, May 2007.
[59] S. I. Barr and J. L. Bedford, “Body mass index, eating attitudes and behaviors, body image and physical activity in young Caucasian and Asian Canadian women,” *FASEB J.*, vol. 23, Apr. 2009.
[60] J. WARDLE, R. BINDRA, B. FAIRCLOUGH, and A. WESTCOMBE, “CULTURE AND BODY-IMAGE - BODY PERCEPTION AND WEIGHT CONCERN IN YOUNG ASIAN AND CAUCASIAN BRITISH WOMEN,” *J. Community Appl. Soc. Psychol.*, vol. 3, no. 3, pp. 173–181, Aug. 1993.
[61] J. L. Bedford and C. S. Johnson, “Societal influences on body image dissatisfaction in younger and older women,” *J. Women Aging*, vol. 18, no. 1, pp. 41–55, 2006.
[62] L. Cevnik, “THE IMPACT OF VALUE ORIENTATIONS ON BODY IMAGE AND THE USE OF PHYSICAL BEAUTY PRACTICES BY YOUNG WOMEN IN SLOVENIA, CROATIA AND SERBIA,” *SOCIOLOGIJA*, vol. 58, no. 1, pp. 53–71, 2016.
[63] D. J. Anschutz and R. C. M. E. Engels, “The Effects of Playing with Thin Dolls on Body Image and Food Intake in Young Girls,” *Sex Roles*, vol. 63, no. 9–10, pp. 621–630, Nov. 2010.
[64] R. C. J. Hermans, J. K. Larsen, C. P. Herman, and R. C. M. E. Engels, “Modeling of palatable food intake in female young adults. Effects of perceived body size,” *Appetite*, vol. 51, no. 3, pp. 512–518, Nov. 2008.
[65] H. M. Al-Hourani and M. F. Atoum, “Body composition, nutrient intake and physical activity patterns in young women during Ramadan,” *Singapore Med. J.*, vol. 48, no. 10, pp. 906–910, Oct. 2007.
[66] K. Murakami, S. Sasaki, H. Okubo, Y. Takahashi, Y. Hosoi, and M. Itabashi, “Monetary costs of

dietary energy reported by young Japanese women: association with food and nutrient intake and body mass index," *Public Health Nutr.*, vol. 10, no. 12, pp. 1430–1439, Dec. 2007.

[67] R. Abed *et al.*, "Eating Disorders and Intrasexual Competition: Testing an Evolutionary Hypothesis among Young Women," *Sci. WORLD J.*, 2012.

[68] V. M. Quick, R. McWilliams, and C. Byrd-Bredbenner, "Fatty, Fatty, Two-by-Four: Weight-Teasing History and Disturbed Eating in Young Adult Women," *Am. J. Public Health*, vol. 103, no. 3, pp. 508–515, Mar. 2013.

[69] A. B. Archibald, J. A. Graber, and J. Brooks-Gunn, "Associations among parent-adolescent relationships, pubertal growth, dieting, and body image in young adolescent girls: A short-term longitudinal study," *J. Res. Adolesc.*, vol. 9, no. 4, pp. 395–415, 1999.

[70] E. Taniguchi and C. M. Thompson, "Family Communication and Body Dissatisfaction among Young Adult Females: Mediating Effects of Social Competence," *J. Fam. Commun.*, vol. 15, no. 4, pp. 410–427, 2015.

[71] A. Nerini, C. Matera, and C. Stefanile, "Siblings' appearance-related commentary, body dissatisfaction, and risky eating behaviors in young women," *Eur. Rev. Appl. Psychol. Eur. Psychol. Appl.*, vol. 66, no. 6, pp. 269–276, Nov. 2016.

[72] D. L. Oswald, S. L. Franzoi, and K. A. Frost, "EXPERIENCING SEXISM AND YOUNG WOMEN'S BODY ESTEEM," *J. Soc. Clin. Psychol.*, vol. 31, no. 10, pp. 1112–1137, Dec. 2012.

[73] D. McGrane and A. Carr, "Young women at risk for eating disorders: Perceived family dysfunction and parental psychological problems," *Contemp. Fam. Ther.*, vol. 24, no. 2, pp. 385–395, Jun. 2002.

[74] D. S. Abebe, L. Torgersen, L. Lien, G. S. Hafstad, and T. von Soest, "Predictors of disordered eating in adolescence and young adulthood A population-based, longitudinal study of females and males in Norway," *Int. J. Behav. Dev.*, vol. 38, no. 2, pp. 128–138, Mar. 2014.

[75] S. S. Brady, "Lifetime family violence exposure is associated with current symptoms of eating disorders among both young men and women," *J. Trauma. Stress*, vol. 21, no. 3, pp. 347–351, Jun. 2008.

[76] A. Favaro, S. Ferrara, and P. Santonastaso, "The spectrum of eating disorders in young women: A prevalence study in a general population sample," *Psychosom. Med.*, vol. 65, no. 4, pp. 701–708, 2003.

[77] D. J. Moorhead, C. K. Stashwick, H. Z. Reinherz, R. M. Giaconia, R. M. Streigel-Moore, and A. D. Paradis, "Child and adolescent predictors for eating disorders in a community population of young adult women," *Int. J. Eat. Disord.*, vol. 33, no. 1, pp. 1–9, Jan. 2003.

[78] E. T. Barker and N. L. Galambos, "Body dissatisfaction, living away from parents, and poor social adjustment predict binge eating symptoms in young women making the transition to university," *J. Youth Adolesc.*, vol. 36, no. 7, pp. 904–911, Oct. 2007.

[79] J. Ferdos and M. M. Rahman, "Exposure to intimate partner violence and malnutrition among young adult Bangladeshi women: cross-sectional study of a nationally representative sample," *Cad. Saude Publica*, vol. 34, no. 7, 2018.

[80] L. B. Shomaker and W. Furman, "Same-sex peers'influence on young women's body image: An experimental manipulation," *J. Soc. Clin. Psychol.*, vol. 26, no. 8, pp. 871–895, Oct. 2007.

[81] L. Clark and M. Tiggemann, "Sociocultural and individual psychological predictors of body image in young girls: A prospective study," *Dev. Psychol.*, vol. 44, no. 4, pp. 1124–1134, Jul. 2008.

[82] E. Taniguchi and A. S. E. Hubbard, "Effects of Physical Appearance Social Comparisons and Perceived Attainability of an Ideal Body on Body Dissatisfaction and Weight-Management Behaviors

Among Young Japanese Women," *Jpn. Psychol. Res.*

[83] W. Tan *et al.*, "Trait body image flexibility as a predictor of body image states in everyday life of young Australian women," *Body Image*, vol. 30, pp. 212–220, Sep. 2019.

[84] Y. Oruclular and E. Bariskin, "Autonomous-related self, eating attitude and body satisfaction in young females," *Eat. Weight Disord. Anorex. Bulim. Obes.*, vol. 20, no. 3, pp. 337–343, Sep. 2015.

[85] K.-M. Lee, M.-S. Seo, J.-Y. Shim, and Y.-J. Lee, "Body weight status misperception and its association with weight control behaviours, depressive mood and psychological distress in nulliparous normal-weight young women," *Ann. Hum. Biol.*, vol. 42, no. 6, pp. 528–532, 2015.

[86] E. Denisoff and N. S. Endler, "Life experiences, coping, and weight preoccupation in young adult women," *Can. J. Behav. Sci. Can. DES Sci. DU Comport.*, vol. 32, no. 2, pp. 97–103, Apr. 2000.

[87] T. D. Wade, S. M. Wilksch, S. J. Paxton, S. M. Byrne, and S. B. Austin, "How perfectionism and ineffectiveness influence growth of eating disorder risk in young adolescent girls," *Behav. Res. Ther.*, vol. 66, pp. 56–63, Mar. 2015.

[88] A. M. Thompson and K. E. Chad, "The relationship of social physique anxiety to risk for developing an eating disorder in young females," *J. Adolesc. Heal.*, vol. 31, no. 2, pp. 183–189, Aug. 2002.

[89] E. M. COHENTOVEE, "DEPRESSED MOOD AND CONCERN WITH WEIGHT AND SHAPE IN NORMAL YOUNG-WOMEN," *Int. J. Eat. Disord.*, vol. 14, no. 2, pp. 223–227, Sep. 1993.

[90] S. A. Gustafsson, B. Edlund, L. Kjellin, and C. Norring, "Personal Standards, Self-evaluation and Perceived Benefits of Thinness in Girls and Young Women with Disturbed Eating," *Eur. Eat. Disord. Rev.*, vol. 16, no. 6, pp. 463–471, 2008.

[91] G. Dada, G. Feixas, V. Compan, and A. Montesano, "SELF-CONSTRUCTION, COGNITIVE CONFLICTS, AND DISORDERED EATING ATTITUDES IN YOUNG WOMEN," *J. Constr. Psychol.*, vol. 25, no. 1, pp. 70–89, 2012.

[92] T. Purton *et al.*, "Body dissatisfaction, narcissism and self-esteem in young men and women: A moderated mediation analysis," *Pers. Individ. Dif.*, vol. 131, pp. 99–104, Sep. 2018.

[93] C. Dryden and J. Anderson, "The dark triad, trait-based self-objectification, and body image concerns in young women," *Pers. Individ. Dif.*, vol. 145, pp. 1–8, Jul. 2019.

[94] M. J. Jacobs, A. Winzelberg, D. E. Wilfley, and C. B. Taylor, "Spirituality among young women at risk for eating disorders," *Int. J. Eat. Disord.*, vol. 35, no. 4, pp. 436–437, May 2004.

[95] S. L. Weinberger-Litman, L. A. Rabin, J. Fogel, J. L. Mensinger, and L. Litman, "Psychosocial Mediators of the Relationship Between Religious Orientation and Eating Disorder Risk Factors in Young Jewish Women," *Psycholog. Relig. Spiritual.*, vol. 8, no. 4, pp. 265–276, Nov. 2016.

[96] R. Andrew, M. Tiggemann, and L. Clark, "Positive body image and young women's health: Implications for sun protection, cancer screening, weight loss and alcohol consumption behaviours," *J. Health Psychol.*, vol. 21, no. 1, pp. 28–39, Jan. 2016.

[97] I. R. Contento, C. Basch, and P. Zybert, "Body image, weight, and food choices of Latina women and their young children," *J. Nutr. Educ. Behav.*, vol. 35, no. 5, pp. 236–248, 2003.

[98] N. Mori, K. Asakura, and S. Sasaki, "Differential dietary habits among 570 young underweight Japanese women with and without a desire for thinness: a comparison with normal weight counterparts," *Asia Pac. J. Clin. Nutr.*, vol. 25, no. 1, pp. 97–107, 2016.

[99] P. Crocker, N. Kowalski, K. Kowalski, K. Chad, L. Humbert, and S. Forrester, "Smoking behaviour and dietary restraint in young adolescent women: The role of physical self-perceptions,"

Can. J. PUBLIC Heal. Can. SANTE PUBLIQUE, vol. 92, no. 6, pp. 428–432, 2001.

[100] V. Boschi, O. Bellini, G. Matrone, F. Ricciardi Lo Schiavo, and M. Siervo, "Why do normal weight young women look for diet-therapy? Findings from a pilot study in a clinical and non-clinical population," *Eat. Weight Disord.*, vol. 12, no. 2, 2007.

[101] L. M. Ventura, E. T. Randall, J. B. Shapiro, A. C. Kirsch, C. S. Conley, and A. M. Bohnert, "Looking Good and Making It Seem Easy: A Prospective Study of Effortless Perfectionism, Body Image, and BMI in Unhealthy Weight Control Behaviors Among Female Adolescents and Young Adults," *Emerg. ADULTHOOD*, vol. 6, no. 5, pp. 327–335, Oct. 2018.

[102] Z. Abou-Rizk and G. Rail, "``Judging a Body by Its Cover'': Young Lebanese-Canadian Women's Discursive Constructions of the ``Healthy'' Body and ``Health'' Practices," *J. Immigr. Minor. Heal.*, vol. 16, no. 1, pp. 150–164, Feb. 2014.

[103] M. Rahman and A. B. Berenson, "Self-Perception of Weight and Its Association With Weight-Related Behaviors in Young, Reproductive-Aged Women," *Obstet. Gynecol.*, vol. 116, no. 6, pp. 1274–1280, Dec. 2010.

[104] A. J. Gruber, H. G. Pope, J. K. Lalonde, and J. I. Hudson, "Why do young women diet? The roles of body fat, body perception, and body ideal," *J. Clin. Psychiatry*, vol. 62, no. 8, pp. 609–611, Aug. 2001.

[105] S. Lee, J. Suk, M. Kim, and H. Hwang, "How does young women's weight perception affect their nutrient intake?: Korean young women's dietary consumption and nutritional deficiency," *Health Care Women Int.*, vol. 38, no. 11, pp. 1247–1260, 2017.

[106] S. F. Abraham, "Dieting, body weight, body image and self-esteem in young women: doctors' dilemmas," *Med. J. Aust.*, vol. 178, no. 12, pp. 607–611, Jun. 2003.

[107] A. Dakanalis *et al.*, "Risk and maintenance factors for young women's DSM-5 eating disorders," *Arch. WOMENS Ment. Heal.*, vol. 20, no. 6, pp. 721–731, Dec. 2017.

[108] A. Dakanalis, M. A. Zanetti, G. Riva, and M. Clerici, "Psychosocial moderators of the relationship between body dissatisfaction and symptoms of eating disorders: A look at a sample of young Italian women," *Eur. Rev. Appl. Psychol. Eur. Psychol. Appl.*, vol. 63, no. 5, pp. 323–334, Sep. 2013.

[109] N. L.-W. Soh *et al.*, "Body image disturbance in young North European and East Asian women with and without eating disorders in Australia and in Singapore," *Eur. Eat. Disord. Rev.*, vol. 16, no. 4, pp. 287–296, 2008.

[110] S. Trainer, "Negotiating weight and body image in the UAE: Strategies among young emirati women," *Am. J. Hum. Biol.*, vol. 24, no. 3, pp. 314–324, 2012.

[111] A. Conley and J. D. Boardman, "Weight overestimation as an indicator of disordered eating behaviors among young women in the United States," *Int. J. Eat. Disord.*, vol. 40, no. 5, pp. 441–445, Jul. 2007.

[112] E. F. Mantilla and A. Birgegard, "The enemy within: the association between self-image and eating disorder symptoms in healthy, non help-seeking and clinical young women," *J. Eat. Disord.*, vol. 3, 2015.

[113] S. J. Paxton and A. Sculthorpe, "Disordered eating and sex role characteristics in young women: Implications for sociocultural theories of disturbed eating," *Sex Roles*, vol. 24, no. 9–10, pp. 587–598, May 1991.

[114] E. Boujut and K. Gana, "Relationship between depressive mood and eating disorders in a non-clinical young female sample: A one-year longitudinal analysis of cross-lagged and simultaneous

effects," *Eat. Behav.*, vol. 15, no. 3, pp. 434–440, Aug. 2014.

[115] J. Chen *et al.*, "Negative Affect Mediates Effects of Psychological Stress on Disordered Eating in Young Chinese Women," *PLoS One*, vol. 7, no. 10, Oct. 2012.

[116] I. Katsounari, "Self-Esteem, Depression and Eating Disordered Attitudes: A Cross-Cultural Comparison Between Cypriot and British Young Women," *Eur. Eat. Disord. Rev.*, vol. 17, no. 6, pp. 455–461, 2009.

[117] L. Evans, G. A. Kennedy, and E. H. Wertheim, "An examination of the association between eating problems, negative mood, weight and sleeping quality in young women and men," *Eat. Weight Disord.*, vol. 10, no. 4, pp. 245–250, 2005.

[118] K. Ball, C. Lee, and W. Brown, "Psychological stress and disordered eating: An exploratory study with young Australian women," *Women Health*, vol. 29, no. 2, pp. 1–15, 1999.

[119] J. L. Cooper, A. E. O'Shea, M. J. Atkinson, and T. D. Wade, "Examination of the Difficulties in Emotion Regulation Scale and Its Relation to Disordered Eating in a Young Female Sample," *Int. J. Eat. Disord.*, vol. 47, no. 6, pp. 630–639, Sep. 2014.

[120] J. L. Cooper and T. D. Wade, "The Relationship Between Memory and Interpretation Biases, Difficulties with Emotion Regulation, and Disordered Eating in Young Women," *Cognit. Ther. Res.*, vol. 39, no. 6, pp. 853–862, Dec. 2015.

[121] V. Costarelli, M. Demerzi, and D. Stamou, "Disordered eating attitudes in relation to body image and emotional intelligence in young women," *J. Hum. Nutr. Diet.*, vol. 22, no. 3, pp. 239–245, Jun. 2009.

[122] A. Hernando, R. Pallas, A. Cebolla, J. Garcia-Campayo, C. J. Hoogendoorn, and J. Francisco Roy, "Mindfulness, rumination, and coping skills in young women with Eating Disorders: A comparative study with healthy controls," *PLoS One*, vol. 14, no. 3, Mar. 2019.

[123] C. P. Butkowski, T. L. Dixon, and K. Weeks, "Body Surveillance on Instagram: Examining the Role of Selfie Feedback Investment in Young Adult Women's Body Image Concerns," *Sex Roles*, vol. 81, no. 5–6, pp. 385–397, Sep. 2019.

[124] J. Mond, T. C. Myers, R. Crosby, P. Hay, and J. Mitche, "`Excessive exercise' and eating-disordered behaviour in young adult women: Further evidence from a primary care sample," *Eur. Eat. Disord. Rev.*, vol. 16, no. 3, pp. 215–221, 2008.

[125] G. Fouts and K. Vaughan, "Locus of control, television viewing, and eating disorder symptomatology in young females," *J. Adolesc.*, vol. 25, no. 3, pp. 307–311, Jun. 2002.

[126] L. Rothschild-Yakar, M. Peled, A. Enoch-Levy, E. Gur, and D. Stein, "``Eating Me Up from Inside{''}: A Pilot Study of Mentalization of Self and Others and Emotion Regulation Strategies among Young Women with Eating Disorders," *Isr. J. Psychiatry Relat. Sci.*, vol. 55, no. 1, pp. 35–44, 2018.

[127] J. A. Levinson, V. Sarda, K. Sonneville, J. P. Calzo, S. Ambwani, and S. B. Austin, "Diet Pill and Laxative Use for Weight Control and Subsequent Incident Eating Disorder in US Young Women: 2001-2016," *Am. J. Public Health*, vol. 110, no. 1, pp. 109–111, Jan. 2020.

[128] J. Verstuyf, M. Vansteenkiste, B. Soetens, and B. Soenens, "Motivational dynamics underlying eating regulation in young and adult female dieters: relationships with healthy eating behaviours and disordered eating symptoms," *Psychol. Health*, vol. 31, no. 6, pp. 711–729, 2016.

[129] L. Taranis and C. Meyer, "Associations Between Specific Components of Compulsive Exercise and Eating-Disordered Cognitions and Behaviors Among Young Women," *Int. J. Eat. Disord.*, vol. 44, no. 5, pp. 452–458, Jul. 2011.

[130] B. W. Bailey, P. Borup, J. D. LeCheminant, L. A. Tucker, and J. Bromley, "Examining the Relationship Between Physical Activity Intensity and Adiposity in Young Women," *J. Phys. Act. Health*, vol. 12, no. 6, pp. 764–769, Jun. 2015.

[131] Y. Ibe *et al.*, "Association of eating three meals irregularly with changes in BMI and weight among young Japanese men and women: A 2-year follow-up," *Physiol. Behav.*, vol. 163, pp. 81–87, Sep. 2016.

[132] T. M. Dunkel, D. Davidson, and S. Qurashi, "Body satisfaction and pressure to be thin in younger and older Muslim and non-Muslim women: The role of Western and non-Western dress preferences," *Body Image*, vol. 7, no. 1, pp. 56–65, Jan. 2010.

[133] M. L. DeLeel, T. L. Hughes, J. A. Miller, A. Hipwell, and L. A. Theodore, "PREVALENCE OF EATING DISTURBANCE AND BODY IMAGE DISSATISFACTION IN YOUNG GIRLS: AN EXAMINATION OF THE VARIANCE ACROSS RACIAL AND SOCIOECONOMIC GROUPS," *Psychol. Sch.*, vol. 46, no. 8, pp. 767–775, Sep. 2009.

[134] A.-M. Polimeni, S. B. Austin, and A. M. Kavanagh, "Sexual Orientation and Weight, Body Image, and Weight Control Practices among Young Australian Women," *J. WOMENS Heal.*, vol. 18, no. 3, pp. 355–362, Mar. 2009.

[135] N. Nagai, N. Sakane, K. Kotani, T. Hamada, K. Tsuzaki, and T. Moritani, "Uncoupling protein 1 gene-3826 A/G polymorphism is associated with weight loss on a short-term, controlled-energy diet in young women," *Nutr. Res.*, vol. 31, no. 4, pp. 255–261, Apr. 2011.

[136] C. DAVIS, J. DURNIN, M. GUREVICH, A. LEMAIRE, and M. DIONNE, "BODY-COMPOSITION CORRELATES OF WEIGHT DISSATISFACTION AND DIETARY RESTRAINT IN YOUNG-WOMEN," *Appetite*, vol. 20, no. 3, pp. 197–207, Jun. 1993.

[137] K. E. ufemi. Przybyłowicz, D. Jesiołowska, M. Obara-Gołębiowska, and L. Antoniak, "A subjective dissatisfaction with body weight in young women: do eating behaviours play a role?," *Rocz. Państwowego Zakładu Hig.*, vol. 65, no. 3, pp. 243–249, 2014.

[138] N. M. Riley *et al.*, "Relation of self-image to body size and weight loss attempts in black women: the CARDIA study. Coronary Artery Risk Development in Young Adults.," *Am. J. Epidemiol.*, vol. 148, no. 11, pp. 1062–1068, 1998.

[139] G. Soohinda, D. Mishra, H. Sampath, and S. Dutta, "Body dissatisfaction and its relation to Big Five personality factors and self-esteem in young adult college women in India," *Indian J. Psychiatry*, vol. 61, no. 4, pp. 400–404, 2019.

[140] M. Balfe, F. Doyle, D. Smith, S. Sreenan, R. Conroy, and R. Brugha, "Dealing with the devil: weight loss concerns in young adult women with type 1 diabetes," *J. Clin. Nurs.*, vol. 22, no. 13–14, pp. 2030–2038, Jul. 2013.

[141] D. Daneman, M. Olmsted, A. Rydall, S. Maharaj, and G. Rodin, "Eating disorders in young women with type 1 diabetes - Prevalence, problems and prevention," *Horm. Res.*, vol. 50, no. 1, pp. 79–86, 1998.

[142] D. Daneman and G. Rodin, "Eating disorders in young women with type 1 diabetes: a cause for concern?," *ACTA Paediatr.*, vol. 88, no. 2, pp. 117–119, Feb. 1999.

[143] R. Cohen, J. Fardouly, T. Newton-John, and A. Slater, "#BoPo on Instagram: An experimental investigation of the effects of viewing body positive content on young women's mood and body image," *NEW MEDIA Soc.*, vol. 21, no. 7, pp. 1546–1564, Jul. 2019.

[144] W. McMillan, E. Stice, and P. Rohde, "High- and Low-Level Dissonance-Based Eating Disorder Prevention Programs With Young Women With Body Image Concerns: An Experimental

Trial," *J. Consult. Clin. Psychol.*, vol. 79, no. 1, pp. 129–134, Feb. 2011.

[145] M. J. Atkinson and T. D. Wade, "Does mindfulness have potential in eating disorders prevention? A preliminary controlled trial with young adult women," *Early Interv. Psychiatry*, vol. 10, no. 3, pp. 234–245, Jun. 2016.

[146] K. A. M. Ginis, H. A. Strong, S. M. Arent, S. R. Bray, and R. L. Bassett-Gunter, "The effects of aerobic- versus strength-training on body image among young women with pre-existing body image concerns," *Body Image*, vol. 11, no. 3, pp. 219–227, Jun. 2014.

[147] A. Osei *et al.*, "Combining Home Garden, Poultry, and Nutrition Education Program Targeted to Families With Young Children Improved Anemia Among Children and Anemia and Underweight Among Nonpregnant Women in Nepal," *FOOD Nutr. Bull.*, vol. 38, no. 1, pp. 49–64, Mar. 2017.

[148] H.-A. Scott, M. Marcinow, R. Hicks, and A. C. Buchholz, "Breakfast Cereal Consumption Moderates the Association Between Body Composition and Body Esteem in Young Women but not in Young Men," *Can. J. Diet. Pract. Res.*, vol. 75, no. 4, pp. 214–217, Dec. 2014.

[149] A. Slater, N. Cole, and J. Fardouly, "The effect of exposure to parodies of thin-ideal images on young women's body image and mood," *Body Image*, vol. 29, pp. 82–89, Jun. 2019.

[150] J. L. Stapleton, S. L. Manne, A. K. Day, K. Levonyan-Radloff, and S. L. Pagoto, "Healthy Body Image Intervention Delivered to Young Women via Facebook Groups: Formative Study of Engagement and Acceptability," *JMIR Res. Protoc.*, vol. 7, no. 2, Feb. 2018.

[151] E. Stice, P. Rohde, S. Durant, and H. Shaw, "A Preliminary Trial of a Prototype Internet Dissonance-Based Eating Disorder Prevention Program for Young Women With Body Image Concerns," *J. Consult. Clin. Psychol.*, vol. 80, no. 5, pp. 907–916, Oct. 2012.

[152] A. M. Toole and L. W. Craighead, "Brief self-compassion meditation training for body image distress in young adult women," *Body Image*, vol. 19, pp. 104–112, Dec. 2016.

[153] E. R. Boyd and J. Moncrieff-Boyd, "Swimsuit issues: promoting positive body image in young women's magazines," *Heal. Promot. J. Aust.*, vol. 22, no. 2, pp. 102–106, Aug. 2011.

[154] S. M. Wilksch, A. O'Shea, C. B. Taylor, D. Wilfley, C. Jacobi, and T. D. Wade, "Online prevention of disordered eating in at-risk young-adult women: a two-country pragmatic randomized controlled trial," *Psychol. Med.*, vol. 48, no. 12, pp. 2034–2044, Sep. 2018.